

POSTPHARYNGEAL FOREGUT

The primitive gut is divided by head- and tail-folding into three main compartments. The foregut extends from the buccopharyngeal membrane to its continuation into the central midgut region via the cranial intestinal portal. The midgut extends between the intestinal portals and, in the early embryo, is in wide communication with the yolk sac. The hindgut extends from the caudal intestinal portal to the cloacal membrane. The cranial end of the foregut, the embryonic pharynx, is intimately associated with head and neck development (Ch. 36). The portion of foregut that passes dorsal to the pericardial cavity gives rise to the respiratory diverticulum and oesophagus within the thorax (Chs 36 and 52). Caudal to the developing diaphragm, the enteric gut is conventionally subdivided into three embryological portions: fore-, mid- and hindgut. There are no corresponding fundamental morphological and cytological distinctions between the three parts (Fig. 60.1), and so the foregut produces a portion of the duodenum, as does the midgut, and the midgut similarly produces large intestine, as does the hindgut. The differences between the portions of the gut develop as a result of interactions between the three embryonic tissue layers that give rise to the gut: namely, the endodermal inner epithelium, the thick layer of splanchnopleuric mesenchyme, and the outer layer of proliferating splanchnopleuric coelomic epithelium.

The epithelial layer of the mucosa and connected ducts and glands are derived from the endodermal epithelium. The lamina propria and

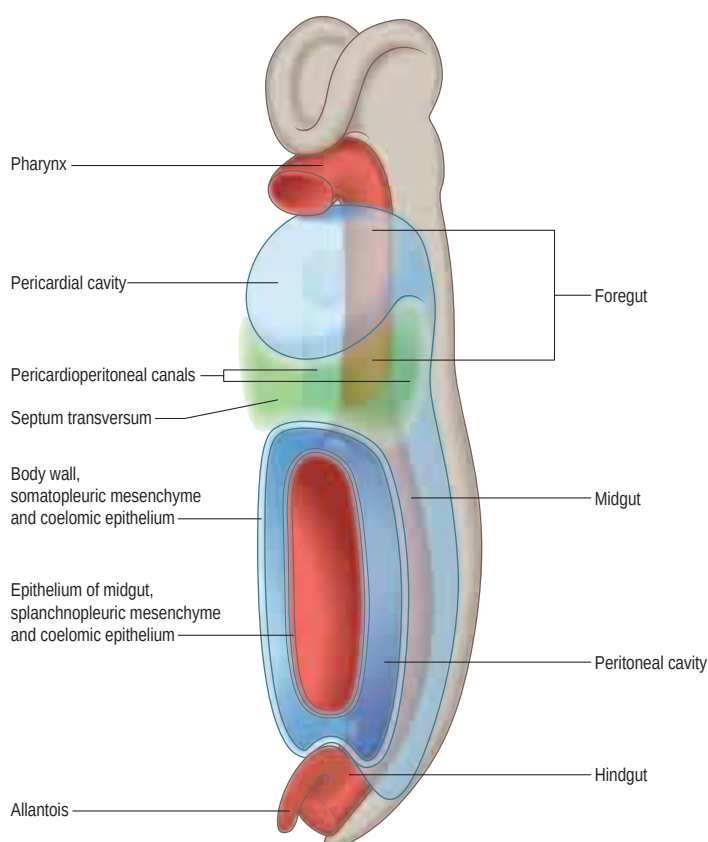


Fig. 60.1 Major epithelial populations within the early embryo. The early gut tube is close to the notochord and neural tube dorsally. The splanchnopleuric layer of the intraembryonic coelomic epithelium is in contact with the foregut ventrally and laterally, with the midgut laterally, and the hindgut ventrally and laterally.

muscularis mucosa, the connective tissue of the submucosa, the muscularis externa and the external connective tissue are all derived from the splanchnopleuric mesenchyme. The outer peritoneal epithelium is derived from the splanchnopleuric coelomic epithelium.

Throughout the gut, blood vessels, lymphatics and lymph nodes develop from local populations of angiogenic mesenchyme. The nerves, which are distributed within the enteric and autonomic systems, are derived from the neural crest. There is a craniocaudal developmental gradient along the gut, in that the stomach and small intestine develop in advance of the colon.

Figure 60.1 shows the fundamental relationship of the intraembryonic coelom to the developing gut. Figure 60.2 shows the gut in a stage 12 embryo in relation to the other developing viscera, especially the heart and liver. Figure 60.3 shows the overall development of the gut from stages 13 to 17. These diagrams should be compared.

All regions of the gut develop from epithelial-mesenchymal interactions that are dependent on the sequential expression of a range of basic and specific genes; on the regulation of the developmental clock, seen in all areas of development; and on endogenous regulatory mechanisms and local environmental influences (Lebenthal 1989). Although all these factors pertain to the whole range of developing tissues, local differences in any one of these factors along the length of the developing gut promote the differentiation of, for example, the gastric mucosa and hepatocytes; the rotation of the midgut; and the final disposition of the sessile portions of the fully formed gastrointestinal tract. The hedgehog (Hh) ligands, Shh, Ihh and Dhh are expressed in the developing gut: Shh and Ihh in the endodermal epithelium and Dhh in endothelial cells. These ligands bind to the Patched receptors (Ptc-1 and Ptc-2), which activate the transcription factor Gli3. Knockout of Shh and Ihh has been associated with oesophageal atresia, gut malrotation, decreased development of the muscularis propria, enteric neurone anomalies and imperforate anus (Kolterud et al 2009). The gut is functional prior to birth and able to interact with the extrauterine environment in preterm infants.

OESOPHAGUS

The oesophagus can be distinguished from the stomach at stage 13 (embryo length 5 mm). It elongates during successive stages and its absolute length increases more rapidly than the embryo as a whole. Cranially, it is invested by splanchnopleuric mesenchyme posterior to the developing trachea and, more caudally, between the developing lungs and pericardioperitoneal canals posterior to the pericardium. Caudal to the pericardium, the terminal, pregastric segment of the oesophagus has a short, thick, dorsal meso-oesophagus (from splanchnopleuric mesenchyme), while ventrally, it is enclosed in the cranial stratum of the septum transversum mesenchyme. Each of the above is continuous caudally with its respective primitive dorsal and ventral mesogastria. Thus, the oesophagus has only limited areas related to a primary coelomic epithelium. However, it is important to note the subsequent development of the para-oesophageal right and left pneumatoenteric recesses (see Fig. 60.7), the relation of the ventral aspect of the middle third of the oesophagus to the oblique sinus of the pericardium, and the relation of its lateral walls in the lower thorax to the mediastinal pleura. All the foregoing are secondary extensions from the primary coelom.

The oesophageal mucosa consists of two layers of cells by stage 15 (week 5), but the proliferation of the mucosa does not occlude the lumen at any time. The mucosa becomes ciliated at 10 weeks, and stratified squamous epithelium is present at the end of the fifth month; occasionally, patches of ciliated epithelium may be present at birth. Circular muscle can be seen at stage 15 but longitudinal muscle has not been identified until stage 21. Neuroblasts can be demonstrated in the early stages; the myenteric plexuses have cholinesterase activity by

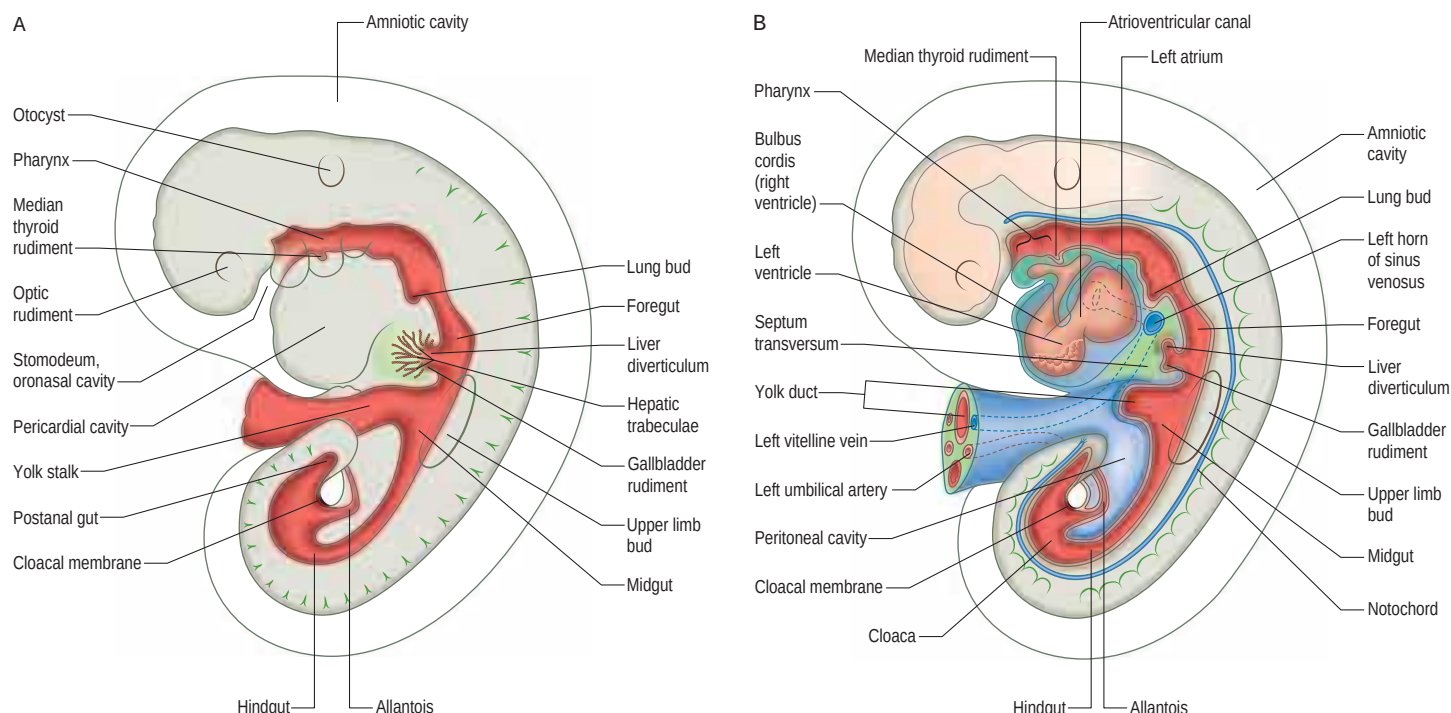


Fig. 60.2 **A**, The digestive tube of a human embryo at stage 12, with 29 paired somites, a crown to rump length of 3.4 mm and an estimated age of 27 days. (Pharyngeal development is followed further in [Figure 36.18](#).) **B**, Reconstruction of a human embryo at the end of the fourth week. The alimentary canal and its outgrowths are shown in median section. The brain is shown in outline but the spinal cord has been omitted. The heart is shown in perspective, and the left horn of the sinus venosus is cut just medial to the entrance of the common cardinal vein (see [Fig. 52.8A–B](#)). The somites are indicated in outline.

9.5 weeks and ganglion cells are differentiated by 13 weeks. It has been suggested that the oesophagus is capable of peristalsis in the first trimester. Periodic fetal swallowing can be seen on ultrasound from 16 weeks. The volume of amniotic fluid ingested increases during the third trimester to more than 500 ml/day. Oesophageal atresia is one of the more common obstructive conditions of the alimentary tract; it may be indicated by polyhydramnios. There is evidence of maturation of the lower oesophageal sphincter at 32 weeks, when, with the prevention of free gastro-oesophageal reflex, stomach size increases ([Hitchcock et al 1992](#)).

Oesophagus at birth

At birth, the oesophagus extends 8–10 cm from the cricoid cartilage to the gastric cardiac orifice. It starts and ends one to two vertebrae, respectively, higher than in the adult, extending from between the fourth to the sixth cervical vertebra to the level of the ninth thoracic vertebra (see [Fig. 14.7](#)). Its average diameter is 5 mm and it possesses the constrictions seen in the adult. The narrowest constriction is at its junction with the pharynx, where the inferior pharyngeal constrictor muscle functions to constrict the lumen; this region may be easily traumatized with instruments or catheters. In the neonate, the mucosa may contain scattered areas of ciliated columnar epithelium but these disappear soon after birth. Peristalsis along the oesophagus and at the lower oesophageal sphincter is immature at birth and results in frequent regurgitation of food in the newborn period. The pressure at the lower oesophageal sphincter approaches that of an adult at 3–6 weeks of age.

STOMACH

At the end of the fourth and beginning of the fifth weeks, the stomach can be recognized as a fusiform dilation cranial to the wide opening of the midgut into the yolk sac (see [Figs 60.2](#) and [60.3](#)). By the fifth week, this opening has narrowed into a tubular vitelline intestinal duct, which soon loses its connection with the digestive tube. At this time, the stomach is median in position and separated cranially from the pericardium by the septum transversum (see [Fig. 60.5A](#)), which extends caudally on to the cranial side of the vitelline intestinal duct and ventrally to the somatopleure. Dorsally, the stomach is related to the aorta and, reflecting the presence of the pleuroperitoneal canals on each side, is connected to the body wall by a short dorsal mesentery, the dorsal mesogastrium (see [Figs 60.5B](#) and [60.6](#)). The latter is directly continu-

ous with the dorsal mesentery (mesenteron) of almost all of the remainder of the intestine, except its caudal short segment.

In human embryos of 10 mm (stages 15–16), the characteristic gastric curvatures are already recognizable. Growth is more active along the dorsal border of the viscus; its convexity markedly increases and the rudimentary fundus appears. Because of more rapid growth along the dorsal border, the pyloric end of the stomach turns ventrally and the concave lesser curvature becomes apparent (see [Figs 60.3](#), [60.6](#)). The stomach is now displaced to the left of the median plane and, apparently, becomes physically rotated, which means that its original right surface becomes dorsal and its left surface becomes ventral. Accordingly, the right vagus is distributed mainly to the dorsal, and the left vagus mainly to the ventral, surfaces of the stomach. The dorsal mesogastrium increases in depth and becomes folded on itself. The ventral mesogastrium becomes more coronal than sagittal. The pancreaticoenteric recess (see [Fig. 60.7B\(ii\)](#)), until this point usually described as a simple depression on the right side of the dorsal mesogastrium, becomes dorsal to the stomach and excavates downwards and to the left between the folded layers. It may now be termed the inferior recess of the bursa omentalis. Put simply, the stomach has undergone two ‘rotations’. The first is 90° clockwise, viewed from the cranial end; the second is 90° clockwise, about an anteroposterior axis. The displacement, morphological changes and apparent ‘rotation’ of the stomach have been attributed variously to its own and surrounding differential growth changes; extension of the pancreaticoenteric recess with changes in its mesenchymal walls; and pressure, particularly that exerted by the rapidly growing liver.

Mucosa

Mucosal and submucosal development can be seen in the eighth to ninth weeks. No villi form in the stomach, unlike in other regions of the gut; instead, glandular pits can be seen in the body and fundus. These develop in the pylorus and cardia by weeks 10 and 11, when parietal cells can be demonstrated. Although acid secretion has not been demonstrated in the fetal stomach before 32 weeks’ gestation, preterm infants, from 26 weeks’ gestation onwards, are able to secrete acid soon after birth. Intrinsic factor has been detected after 11 weeks. This increases from the fourteenth to the twenty-fifth week, at which time the pylorus, which contains more parietal cells than it does in the adult, also contains a relatively larger quantity of intrinsic factor. The significance of the early production of intrinsic factor and the late production of acid by the parietal cells is not known. Chief cells can be

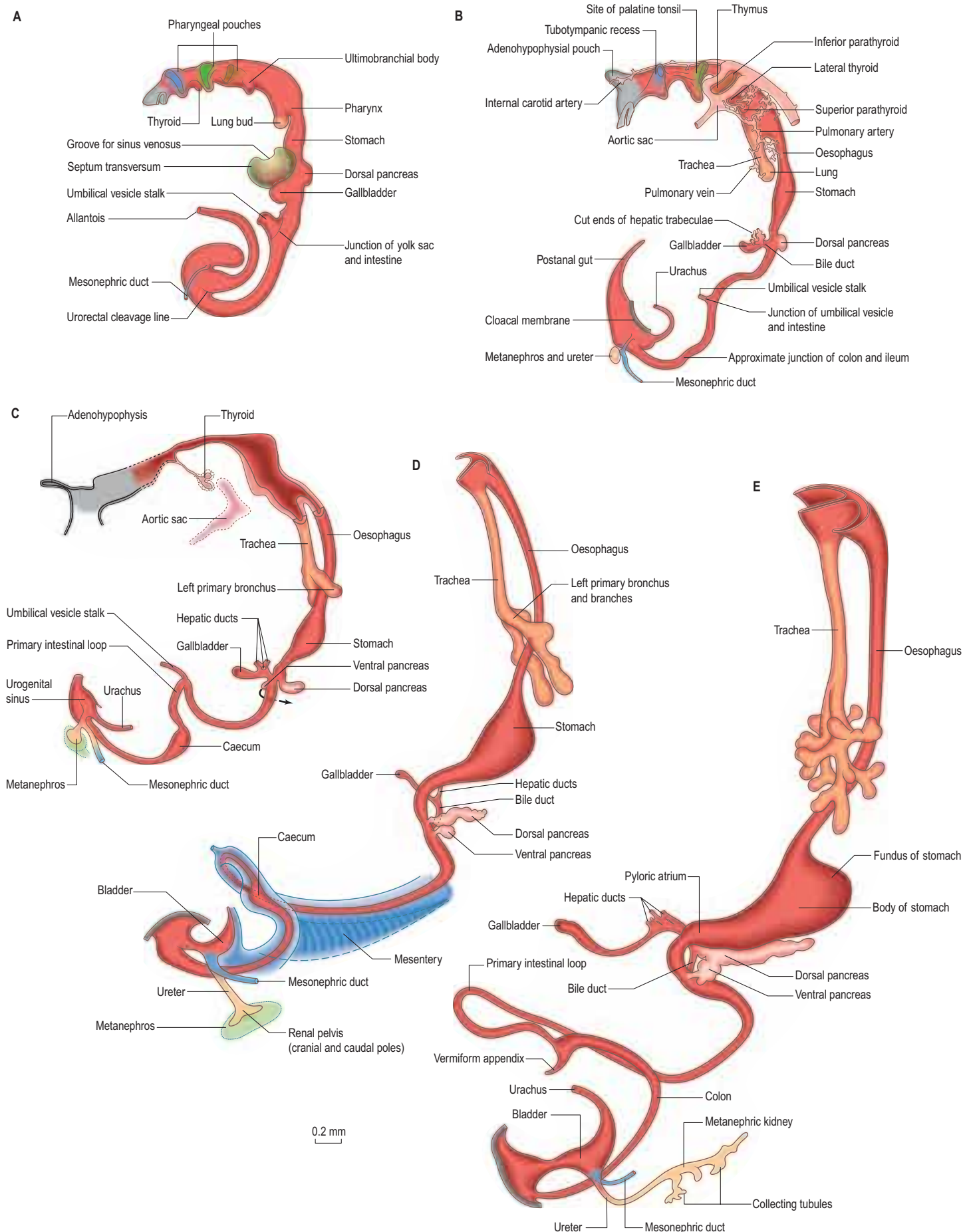


Fig. 60.3 The shape of the endodermal epithelium of the gut at succeeding stages. The scale is constant, illustrating the enormous growth of the gut over a 13-day period. **A**, Stage 13. **B**, Stage 14. **C**, Stage 15. **D**, Stage 16. **E**, Stage 17. Note the separation of the respiratory diverticulum; the elongation of the foregut and expansion of the stomach; the formation of the hepatic and pancreatic diverticula; the lengthening of the midgut loop, which protrudes into the umbilical cord; and the separation of the cloaca into enteric and allantoic portions. (Modified from O'Rahilly and Müller. *Developmental Stages in Human Embryos* 1987 Carnegie Institution of Washington. Pub 637.)

identified after weeks 12–13, although they cannot be demonstrated to contain pepsinogen until term. Mucous neck cells actively produce mucus from week 16. Gastrin-producing cells have been demonstrated in the antrum between 19 and 20 weeks, and gastrin levels have been measured in cord blood and in the plasma at term. Cord serum contains gastrin levels 2–3 times higher than those in maternal serum.

Muscularis

The stomach muscularis externa develops its circular layer at 8–9 weeks, when neural plexuses are developing in the body and fundus. The longitudinal muscle develops slightly later. Few studies note a time of appearance of the oblique layer. The pyloric musculature is thicker than the rest of the stomach; in general, the thickness of the total musculature of the stomach at term is reduced, compared to the adult.

Serosa

The serosa of the stomach is derived from the splanchnopleuric coelomic epithelium. No part of this serosa undergoes absorption. The original left side of the gastric serosa faces the greater sac; the right side faces the lesser sac.

Stomach at birth

The stomach exhibits fetal characteristics until just after birth, when the initiation of pulmonary ventilation, the reflexes of coughing and swallowing, and crying cause the ingestion of large amounts of air and liquid. Once postnatal swallowing has started, the stomach distends to four or five times its contracted state, and shifts its position in relation to the state of expansion and contraction of the other abdominal viscera, and to the position of the body. In the neonate, the anterior surface of the stomach is generally covered by the left lobe of the liver, which extends nearly as far as the spleen (see Fig. 14.6B). Only a small portion of the greater curvature of the stomach is visible anteriorly. The capacity of the stomach is 30–35 ml in the full-term neonate, rising to 75 ml in the second week and 100 ml by the fourth week (adult capacity is, on average, 1000 ml). The mucosa and submucosa are relatively thicker than in the adult; however, the muscularis is only moderately developed and peristalsis is not coordinated. At birth, gastric acid secretion is low, which means that gastric pH is high for the first 12 postnatal hours. It falls rapidly with the onset of gastric acid secretion, usually after the first feed. Acid secretion usually remains low for the first 10 days postnatally. Gastric emptying and transit times are delayed in the neonate.

DUODENUM

The duodenum develops from the caudal part of the foregut and the cranial part of the midgut. A ventral mesoduodenum, which is continuous cranially with the ventral mesogastrium, is attached only to the foregut portion. Posteriorly, the duodenum has a thick dorsal mesoduodenum, which is continuous with the dorsal mesogastrium cranially and the dorsal mesentery of the midgut caudally. Anteriorly, the extreme caudal edge of the ventral mesentery of the foregut extends on to the short initial segment of the duodenum. The liver arises as a diverticulum from the ventral surface of the duodenum at the foregut–midgut junction, i.e. where the midgut is continuous with the yolk sac wall (the cranial intestinal portal). The ventral pancreatic bud also arises from this diverticulum. The dorsal pancreatic bud evaginates posteriorly into the dorsal mesoduodenum slightly more cranially than the hepatic diverticulum. The rotation, differential growth, and cavitations related to the developing stomach and omenta cause corresponding movements in the duodenum, which forms a loop directed to the right, with its original right side now adjacent to the posterior abdominal wall (see Fig. 60.6). This shift is compounded by the migration of the bile duct and ventral pancreatic duct around the duodenal wall. Their origin shifts until it reaches the medial wall of the second part of the fully formed duodenum; the bile duct passes posteriorly to the duodenum and travels in the free edge of the ventral duodenum and ventral mesogastrium. Local adherence and subsequent absorption of part of the duodenal serosa and the parietal peritoneum result in almost the whole of the duodenum, other than a short initial segment, becoming retroperitoneal (sessile).

Duodenal atresia is a developmental defect found in 1 in 5000 live births (Whittle 1999). It may be associated with an annular pancreas, which may compress the duodenum externally (20% of duodenal

atresia), or with anomalies of the bile duct. In 40–60% of cases, the atresia is complete and pancreatic tissue fills the lumen. The condition can be diagnosed on ultrasound examination, which reveals a typical double bubble appearance, caused by fluid enlarging the stomach and the proximal duodenum. Polyhydramnios is invariably present and often is the indication for the scan. Duodenal atresia commonly occurs with other developmental defects, e.g. cardiac and skeletal anomalies, and in Down's syndrome.

DORSAL AND VENTRAL MESENTERIES OF THE FOREGUT

The epithelium of the stomach and duodenum does not rotate relative to its investing mesenchyme. The rotation includes the coelomic epithelial walls of the pericardioperitoneal canals, which are on each side of the stomach and duodenum and form its serosa, and the elongating dorsal mesogastrium or the much shorter dorsal mesoduodenum. A ventral mesogastrium can be seen when the distance between the stomach and liver increases. Whereas the dorsal mesogastrium takes origin from the posterior body wall in the midline, its connection to the greater curvature of the stomach, which lengthens as the stomach grows, becomes directed to the left as the stomach undergoes its first rotation. With the second rotation, a portion of the dorsal mesogastrium now faces caudally (see Fig. 60.6). The ventral mesogastrium remains as a double layer of coelomic epithelium, which encloses mesenchyme and forms the lesser omentum (see Fig. 60.7).

Movement of the stomach is associated with an extensive lengthening of the dorsal mesogastrium, which becomes the greater omentum; this now, from its posterior origin, droops caudally over the small intestine, then folds back anteriorly and ascends to the greater curvature of the stomach. The greater omentum is, therefore, composed of a fold containing, technically, four layers of peritoneum. The dorsal mesoduodenum, or suspensory ligament of the duodenum, is a much thicker structure, and it fixes the position of the duodenum when the rest of the midgut and its dorsal mesentery elongate and pass into the umbilical cord.

SPECIAL GLANDS OF THE POSTPHARYNGEAL FOREGUT

Pancreas

The pancreas develops from two evaginations of the foregut that fuse to form a single organ. A dorsal pancreatic bud can be seen in stage 13 embryos as a thickening of the endodermal tube that proliferates into the dorsal mesogastrium (see Fig. 60.3; Fig. 60.4). A ventral pancreatic bud evaginates in close proximity to the liver primordium but cannot be clearly identified until stage 14, when it appears as an evagination of the bile duct itself. At stage 16 (5 weeks), differential growth of the wall of the duodenum results in movement of the ventral pancreatic bud and the bile duct to the right side and, ultimately, to a dorsal position. It is not clear whether there is a corresponding shift of mesenchyme during this rotation. However, the ventral pancreatic bud and the bile duct rotate from a position within the ventral mesogastrium (ventral mesoduodenum) to one in the dorsal mesogastrium (dorsal mesoduodenum), which is destined to become fixed on to the posterior abdominal wall. By stage 17, the ventral and dorsal pancreatic buds have fused, although the origin of the ventral bud from the bile duct is still obvious. Three-dimensional reconstruction of the ventral and dorsal pancreatic buds has confirmed that the dorsal pancreatic bud forms the anterior part of the head, the body and the tail of the pancreas, and the ventral pancreatic bud forms the posterior part of the head and the posterior part of the uncinate process. The ventral pancreatic bud does not form all of the uncinate process (Collins 2002a).

The developing pancreatic ducts usually fuse in such a way that most of the dorsal duct drains into the proximal part of the ventral duct (see Figs 60.3–60.4). The proximal portion of the dorsal duct usually persists as an accessory duct. The fusion of the ducts takes place late in development or in the postnatal period; 85% of infants have patent accessory ducts, as compared to 40% of adults. Fusion may not occur in 10% of individuals, in which case separate drainage into the duodenum is maintained: so-called pancreatic divisum (pancreas divisum). Failure of the ventral pancreatic diverticulum to migrate will result in an annular pancreas, which may constrict the duodenum locally.

The ventral pancreas does not always extend anterior to the superior mesenteric vein but remains related to its right lateral surface. Initially,

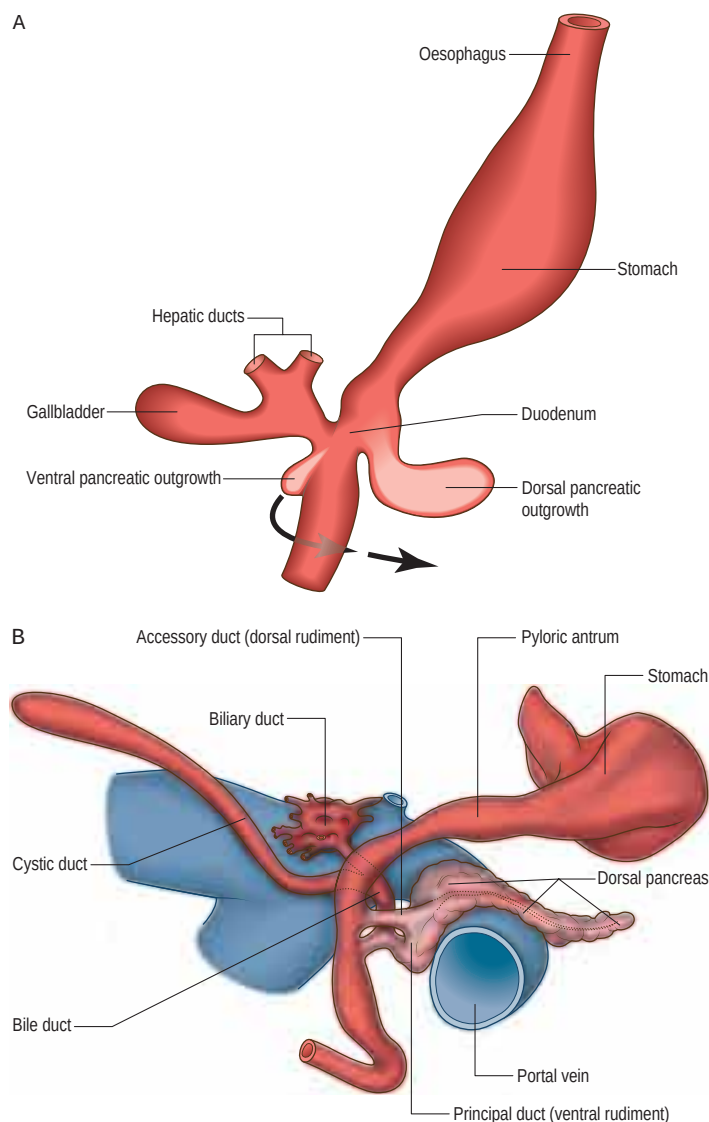


Fig. 60.4 Development of the pancreas in a human embryo. **A**, An early stage, 7.5 mm embryo: lateral view. **B**, A later stage, 14.5 mm embryo: ventral view.

the body of the pancreas extends into the dorsal mesoduodenum and then cranially into the dorsal mesogastrum. As the stomach rotates, this portion of the dorsal mesogastrum is directed to the left, forming the posterior wall of the lesser sac. The posterior layer of this portion of dorsal mesogastrum fuses with the parietal layer of the coelom wall (peritoneum), and the pancreas becomes mainly retroperitoneal (see Fig. 60.7C). The region of fusion of the dorsal mesogastrum does not extend so far left as to include the tail of the pancreas, which passes into the splenorenal (lienorenal) ligament. The anterior border of the pancreas later provides the main line of attachment for the posterior leaves of the greater omentum.

Cellular development of the pancreas

The early specification of pancreatic endoderm involves the proximity of the notochord to the dorsal endoderm, which locally represses the expression of Shh transcription factor. Endoderm caudal to the pancreatic region does not respond to notochordal signals. The ventral pancreatic endoderm does not seem to undergo the same induction. Pancreatic mesenchyme is derived from two regions. The mesenchyme that surrounds the dorsal pancreatic bud proliferates from the splanchnopleuric coelomic epithelium of the medial walls of the pericardioperitoneal canals, whereas the ventral pancreatic bud is invested by septum transversum mesenchyme and by mesenchyme derived from the lower ventral walls of the pericardioperitoneal canals.

The primitive endodermal ductal epithelium provides the stem cell population for all the secretory cells of the pancreas. Initially, these endocrine cells are located in the duct walls or in buds developing from them; later, they accumulate in pancreatic islets. The remaining primitive duct cells will differentiate into definitive ductal cells. In the fetus, they develop microvilli and cilia but lack the lateral interdigitations seen in the adult. Branches of the main duct become interlobular

ductules, which terminate as blind-ending acini or as tubular, acinar elements.

The ductal branching pattern and acinar structure of the pancreas are determined by the pancreatic mesenchyme. This mesenchyme gives rise to connective tissue between the ducts, which, in the fetus, appears to be important in stimulating pancreatic proliferation and maintaining the relative proportions of acinar, α and β cells during development. It also provides cell lines for smooth muscle within the pancreas. Angiogenic mesenchyme invades the developing gland to produce blood and lymphatic vessels.

The process of islet differentiation is divided into two phases (Collins 2002a). Phase I, characterized by proliferation of polyhormonal cells, occurs from weeks 9–15. Phase II, characterized by differentiation of monohormonal cells, is seen from week 16 onwards. The β cells, producing insulin and amylin, differentiate first, followed by α cells, which produce glucagon. The δ cells, which produce somatostatin, are seen after 30 weeks. The dorsal bud gives rise mostly to α cells, and the ventral bud to most of the pancreatic polypeptide-producing cells. The β cells develop from the duct epithelium throughout development and into the neonatal period. Later, in weeks 10–15, some of the primitive ducts differentiate into acinar cells, in which zymogen granules or acinar cell markers can be detected at 12–16 weeks.

The pancreas in the neonate has all of the normal subdivisions of the adult. The head is proportionately large in the newborn and there is a smooth continuation between the body and the tail. The inferior border of the head of the pancreas is found at the level of the second lumbar vertebra. The body and tail pass cranially and to the left, and the tail is in contact with the spleen (see Fig. 14.6).

Liver

The liver is one of the most precocious embryonic organs and is the main centre for haemopoiesis in the fetus. It develops from an endodermal evagination of the foregut and from septum transversum mesenchyme, which is derived from the proliferating coelomic epithelium in the protocardiac region. The development of the liver is intimately related to the development of the heart. The vitelline veins, succeeded by the umbilical veins passing to the sinus venosus, are disrupted by the enlarging septum transversum to form a hepatic plexus, the forerunner of the hepatic sinusoids. (See Collins (2002b) for a detailed account of hepatic development.)

Early liver development

As the head fold and early intraembryonic coelom form, the ventral parietal wall of the pericardial cavity gives rise to populations of cells termed precardiac or cardiac mesenchyme. Hepatic endoderm is induced to proliferate by this mesenchyme, although all portions of the early heart tube, truncus arteriosus, atria, ventricle, both endocardium and myocardium, have hepatic induction potency that is tissue-specific but not species-specific. As the heart and foregut become separated by the accumulation of the cardiac mesenchyme, the mesenchyme itself is renamed septum transversum. It is seen as a ventral mass, caudal to the heart, which passes dorsally on each side of the developing gut to join the mesenchyme proliferating from the walls of the pericardioperitoneal canals. The majority of the cells within the septum transversum are destined to become hepatic mesenchyme. For details of the molecular signalling of early hepatic development, see Lemaigre (2009).

In the stage 11 embryo, the location of the hepatic endoderm has been identified at the superior boundary of the rostral intestinal portal. By stage 12, the hepatic endodermal primordium is directed ventrally and begins to proliferate as a diverticulum. There are two parts: a caudal part, which will produce the cystic duct and gallbladder; and a cranial part, which forms the liver biliary system (see Fig. 60.3; Fig. 60.5A). The cells start to express liver-specific molecular markers and glycogen storage.

Around the cranial portion of the hepatic diverticulum, the basal lamina is progressively disrupted and individual epithelial cells migrate into the surrounding septum transversum mesenchyme. The previously smooth contour of the diverticulum merges into columnar extensions of endoderm, the epithelial trabeculae, which stimulate the hepatic mesenchymal cells to form blood islands and endothelium. The advance of the endodermal epithelial cells promotes the conversion of progressively more hepatic mesenchyme into endothelium and blood cells, and only a little remains to form the scanty liver capsule and interlobular connective tissue. This invasion by the hepatic epithelium is completed in stage 13, when it approaches the caudal surface of the pericardial cavity, and is separated from it only by a thin lamina of mesenchyme that will give rise to part of the diaphragm.

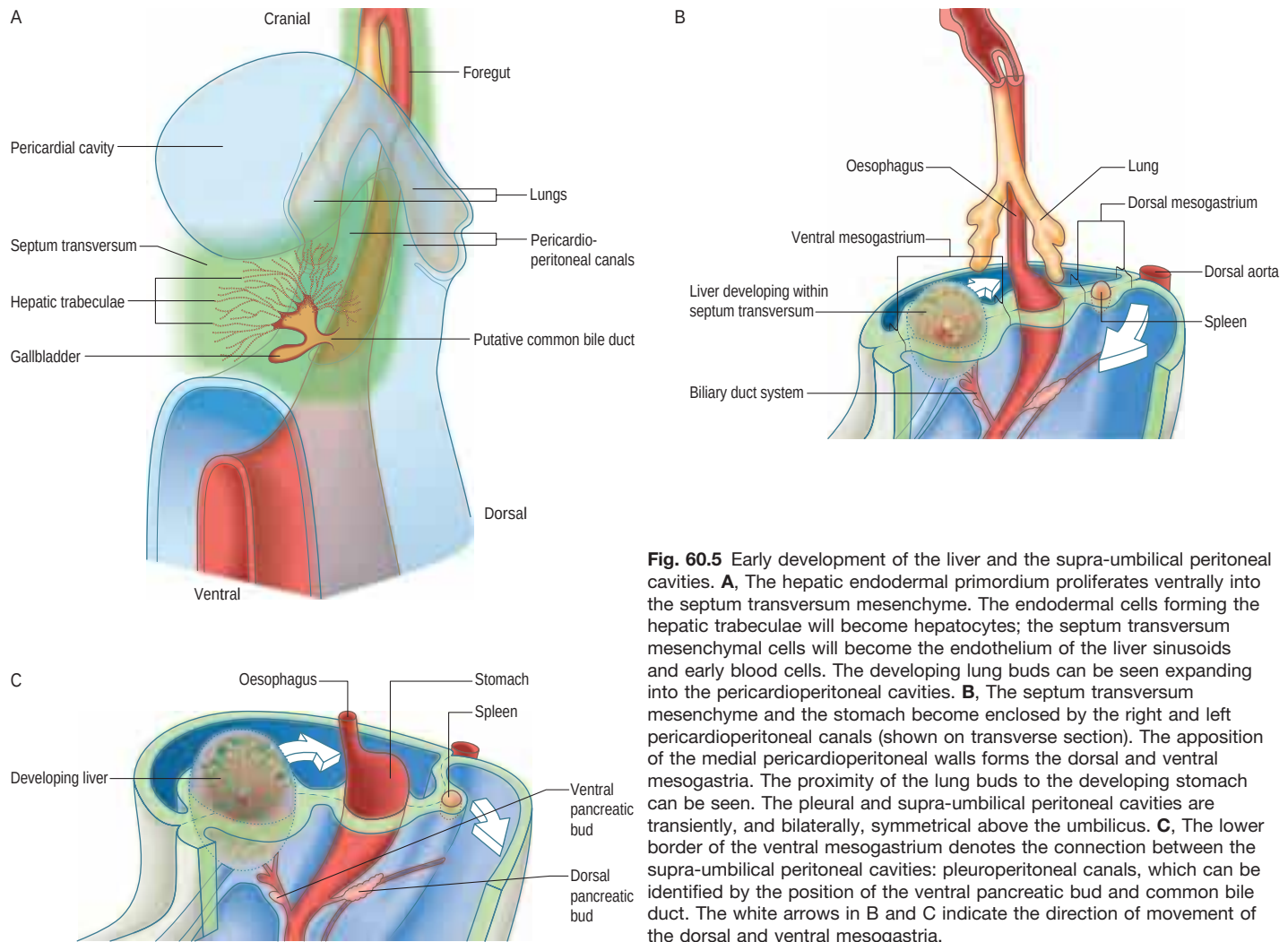


Fig. 60.5 Early development of the liver and the supra-umbilical peritoneal cavities. **A**, The hepatic endodermal primordium proliferates ventrally into the septum transversum mesenchyme. The endodermal cells forming the hepatic trabeculae will become hepatocytes; the septum transversum mesenchymal cells will become the endothelium of the liver sinusoids and early blood cells. The developing lung buds can be seen expanding into the pericardioperitoneal cavities. **B**, The septum transversum mesenchyme and the stomach become enclosed by the right and left pericardioperitoneal canals (shown on transverse section). The apposition of the medial pericardioperitoneal walls forms the dorsal and ventral mesogastria. The proximity of the lung buds to the developing stomach can be seen. The pleural and supra-umbilical peritoneal cavities are transiently, and bilaterally, symmetrical above the umbilicus. **C**, The lower border of the ventral mesogastrium denotes the connection between the supra-umbilical peritoneal cavities: pleuroperitoneal canals, which can be identified by the position of the ventral pancreatic bud and common bile duct. The white arrows in B and C indicate the direction of movement of the dorsal and ventral mesogastria.

During this early phase of development, the liver is far more highly vascularized than the rest of the gut. The hepatic capillary plexus is connected bilaterally with the right and left vitelline veins. Dorsolaterally, they empty by multiple channels into enlarged hepatocardiac channels, which lead to the right and left horns of the sinus venosus (see Fig. 60.9); usually, the channel on the right side is most developed. Both left and right channels bulge into the pericardioperitoneal canals, forming sites for the exchange of fluid from the coelom into the vascular channels. The growth of the hepatic tissue in these regions is sometimes referred to as the left and right horns of the liver.

The liver remains proportionately large during its development and constitutes a sizeable organ dorsal to the heart at stage 14, then more caudally placed by stage 16. By this stage, hepatic ducts can be seen separating the hepatic epithelium from the extrahepatic biliary system, but, even at stage 17, the ducts do not penetrate far into the liver.

Maturation of the liver

At 3 months' gestation, the liver almost fills the abdominal cavity and its left lobe is nearly as large as its right. When the haemopoietic activity of the liver is assumed by the spleen and bone marrow, the left lobe undergoes some degeneration and becomes smaller than the right. The liver remains relatively larger than in the adult throughout the remainder of gestation. In the neonate, it constitutes 4% of the body weight, compared to 2.5–3.5% in adults. It is in contact with the greater part of the diaphragm and extends below the costal margin anteriorly, and, in some cases, to within 1 cm of the iliac crest posteriorly. The left lobe covers much of the anterior surface of the stomach and constitutes nearly one-third of the liver (see Fig. 14.6). Although its haemopoietic functions cease before birth, its enzymatic and synthetic functions are not completely mature at birth. Hepatocytes remain a heterogeneous population with different gene expressions and metabolic functions within different hepatic lobule locations. This metabolic zonation becomes fully established after birth (Lemaigre 2009).

Development of intrahepatic biliary ducts

The development of the intrahepatic biliary ducts follows the branching pattern of the portal vein radicles (Collins 2002b). The cranial hepatic

diverticulum gives rise to the liver hepatocytes, the intrahepatic large bile ducts (right and left hepatic ducts, segmental ducts, area ducts and their first branches) and the small bile ducts (septal bile ducts, interlobular ducts and bile ductules). The portal and hepatic veins arise together from the vitelline veins. Early in development, the accumulation of mesenchyme around these veins is similar, whereas later mesenchyme increases around the portal veins. This is a prerequisite for bile duct development. Primitive hepatocytes surround the portal vein branches and associated mesenchyme, and form a sleeve of cells termed the ductal plate. Individual cells of the ductal plate are termed cholangiocytes. Local hepatoblasts that are adjacent to the cholangiocytes arrange themselves to delineate a bile duct lumen and then also switch to a cholangiocyte lineage (Lemaigre 2009). As the bile ducts develop, angiogenic mesenchymal cells form blood vessels that connect to the hepatic artery from 10 weeks. Thus, the portal triads are patterned by the portal vein radicles, which initially induce bile duct formation and then artery formation. The development of the biliary system extends from the hilum to the periphery. Anomalies of the biliary tree are associated with abnormalities of the branching pattern of the portal vein. The developing bile ducts remain patent throughout development; the solid stage of ductal development previously promulgated has been refuted. Atresia of the extrahepatic bile ducts has been noted, often in association with extrahepatic atresia. The cause of this condition is not clear; inflammatory process may be involved, although some cases have features of ductal plate malformation (Howard 2002).

Development of extrahepatic biliary ducts

The caudal part of the hepatic endodermal diverticulum forms the extrahepatic biliary system, the common hepatic duct, gallbladder, cystic duct and common bile duct.

The bile duct, which originated from the ventral wall of the foregut (now duodenum), migrates with the ventral pancreatic bud, first to the right and then dorsomedially into the dorsal mesoduodenum. The right and left hepatic ducts arise from the cranial end of the common hepatic duct from 12 weeks' gestation.

Atresia of the extrahepatic bile ducts in neonates occurs alone or in conjunction with a range of other anomalies, including situs inversus,

malrotation, polysplenia and cardiac defects. In such cases, the intra-hepatic bile ducts have a mature tubular shape but also show features of ductal plate malformation.

In the neonate, the gallbladder has a smaller peritoneal surface than in the adult, and its fundus often does not extend to the liver margin. It is generally embedded in the liver and, in some cases, may be covered by bands of liver. After the second year, the gallbladder assumes the relative size it has in the adult.

MIDGUT

The midgut forms the third and fourth parts of the duodenum, jejunum, ileum and two-thirds of the way along the transverse colon; its development produces most of the small, and a portion of the large, intestine. In embryos of stages 10 and 11, it extends from the cranial to the caudal intestinal portals and communicates directly with the yolk sac over its entire length. Although it has a dorsal wall, the lateral walls have not yet formed at these stages. By stage 12, the connection with the yolk sac has narrowed, such that the midgut has ventral walls cranially and caudally. This connection is reduced to a yolk stalk containing the vitellointestinal duct during stage 13, at which time the yolk sac appears as a sphere in front of the embryo. Posterior to the midgut, the splanchnopleuric coelomic epithelia converge, forming the dorsal mesentery. Ventrolaterally, the intraembryonic coelom is in wide communication with the extraembryonic coelom. At stage 14, the midgut has increased in length more than the axial length of the embryonic body and, with elongation of the dorsal mesentery, it bulges ventrally, deviating from the median plane. For all these stages, see [Figure 60.3](#).

PRIMARY INTESTINAL (OR MIDGUT) LOOP

The midgut loop can first be seen at stage 15, when a bulge – the caecal bud – can be discerned on the lower limb of the loop, caudal to the yolk stalk (which arises from the apex or summit of the loop). Later, the original proximal limb of the loop moves to the right and the distal limb to the left (see [Fig. 60.3C](#)). The longest portion of the dorsal mesentery is at the level of the yolk stalk; there is less relative lengthening near the caudal end of the duodenum or the cranial half of the colon. The midgut extends into the umbilical coelom, having already rotated through an angle of 90° (anticlockwise, viewed from the ventral aspect). This relative position is approximately maintained so long as the protrusion persists, during which time the proximal limb that forms the small intestine elongates greatly. It becomes coiled, and its adjacent mesentery adopts a pleated appearance. The origin of the root of the mesentery is initially both median and vertical, while, at its intestinal attachment, it is elongated like a ruffle and folded along a horizontal zone. The mesenteric sheet and its contained vessels have spiralled through 90°. The distal, colic, part of the loop elongates less rapidly and has no tendency to become coiled. By the time the fetus has attained a length of 40 mm (10 weeks), the peritoneal cavity has enlarged and the relative size of the liver and mesonephros is much less. The re-entry of the gut occurs rapidly and in a particular sequence, during which it continues the process of rotation. The proximal loop returns first, with the jejunum mainly on the left and the ileum mainly on the right of the subhepatic abdominal cavity. As they re-enter the abdominal cavity, the coils of jejunum and ileum slide inwards over the right aspect of the descending mesocolon, and so displace the descending colon to the left. The transverse colon passes superiorly to the origin of the root of the mesentery ([Fig. 60.6](#)). The caecum is the last to re-enter and, at first, lies on coils of ileum on the right. Later development of the colon leads to its elongation and to the establishment of the hepatic and splenic flexures. A timetable for intestinal rotation in staged human embryos is given by [Kim et al \(2003\)](#).

Anomalies of midgut rotation

If the midgut loop fails to return to the abdominal cavity at the appropriate time, a range of ventral defects can result. Failure of obliteration of the vitellointestinal duct connecting the midgut to the yolk sac results in Meckel's diverticulum. This may present as a short segment of vitelline duct attached to the original ventral side of the ileum; it may remain attached to the umbilicus as a fistula; or it may remain as a ligamentous attachment to the umbilicus.

An umbilical hernia occurs when loops of gut protrude into a widened umbilical cord at term. The degree of protuberance may increase when the infant cries, which raises the intra-abdominal pres-

sure; these hernias usually resolve without treatment. Exomphalos is a ventral wall defect with midline herniation of the intra-abdominal contents into the base of the umbilical cord. Herniated viscera are covered by the peritoneum internally and amnion externally. The omphalocele so formed ranges in size from a large umbilical hernia to a very large mass containing most of the visceral organs. Even after the exomphalos has been repaired, these babies will still have a deficient anterior abdominal wall.

Gastroschisis is a para-umbilical defect of the anterior abdominal wall associated with evisceration of the abdominal organs. The organs are not enclosed in membranes; thus, gastroschisis can be detected by prenatal ultrasonography and differentiated from exomphalos (see [Fig. 14.5B](#)). Gastroschisis is thought to result from periumbilical ischaemia, caused by vascular compromise of either the umbilical vein or arteries. The incidence of this condition appears to be increasing, especially in babies born to young women less than 20 years old ([Whittle 1999](#), [Ionescu et al 2014](#)).

Congenital volvulus arises if the midgut loop does not rotate appropriately. A number of types of this condition are identified. Left-sided colon occurs if the midgut loop has not rotated at all; mixed rotation results in the caecum lying inferior to the pylorus; and failure of appropriate attachment of the peritoneum may result in the small intestine being attached at only two points on the posterior abdominal wall. All of these arrangements lead to a risk of volvulus, which may result in necrosis of the gut.

The position and configuration of the duodenal loop are of particular importance in children. The normal duodenal loop has a U-shaped configuration. The suspensory ligament of the duodenum (ligament of Treitz) is usually found to the left of the body of the first or second lumbar vertebral body after normal gut rotation; any other position of this ligament may indicate some degree of gut malrotation. On barium studies, the duodenojejunal flexure should, thus, lie to the left of the upper lumbar spine at the level of the pylorus.

If the caecum has remained in the right upper quadrant, it may become fixed in that position by peritoneal attachments passing to the right, the so-called Ladd's bands. These may compress the underlying duodenum and give rise to duodenal stenosis. The high positioning of the caecum close to the duodenal jejunal flexure – in some cases, in the midline – is associated with later development of volvulus.

The identification of intestinal malrotation can be made by X-ray investigation. However, ultrasonography has the advantage of showing the position of the superior mesenteric vein and artery. The vein should lie to the right of the artery. Most cases of volvulus will show inversion of this normal relationship but malrotation can occur with apparently normally related vessels, particularly in malrotation with bowel obstruction due to Ladd's bands and not volvulus.

UMBILICAL CORD

During the period when the midgut loop protrudes into the umbilical coelom, the edges of the ventral body wall are becoming relatively closer, forming a more discrete root for the umbilical cord. Somatic mesenchyme, which will form the ventral body wall musculature, migrates into the somatopleuric mesenchyme and passes ventrally towards the midline. The umbilical cord forms all of the ventral body wall between the pericardial bulge and the developing external genitalia. It encloses a portion of the extraembryonic coelom, the umbilical coelom, into which midgut loop protrudes. When the midgut loop is abruptly returned to the abdominal cavity, the more recognizable umbilical cord forms. The vitellointestinal duct and vessels involute. The cranial end of the allantois becomes thinned and its lumen partially obliterated, and it forms the urachus. The mesenchymal core of the umbilical cord is derived by coalescence from somatopleuric amniotic mesenchyme, splanchnopleuric vitellointestinal (yolk sac) mesenchyme and splanchnopleuric allantoic (connecting stalk) mesenchyme. These various layers become fused and are gradually transformed into the viscid, mucoid, connective tissue (Wharton's jelly) that characterizes the more mature cord. The changes in the circulatory system result in a large, cranially orientated, left umbilical vein (the right umbilical vein regresses), and two spirally disposed umbilical arteries (see [Fig. 52.19](#)).

MATURATION OF THE SMALL INTESTINE

Mucosa

The exact timing of the cellular morphogenesis of the gut is difficult to establish, especially as it undergoes a proximodistal gradient in

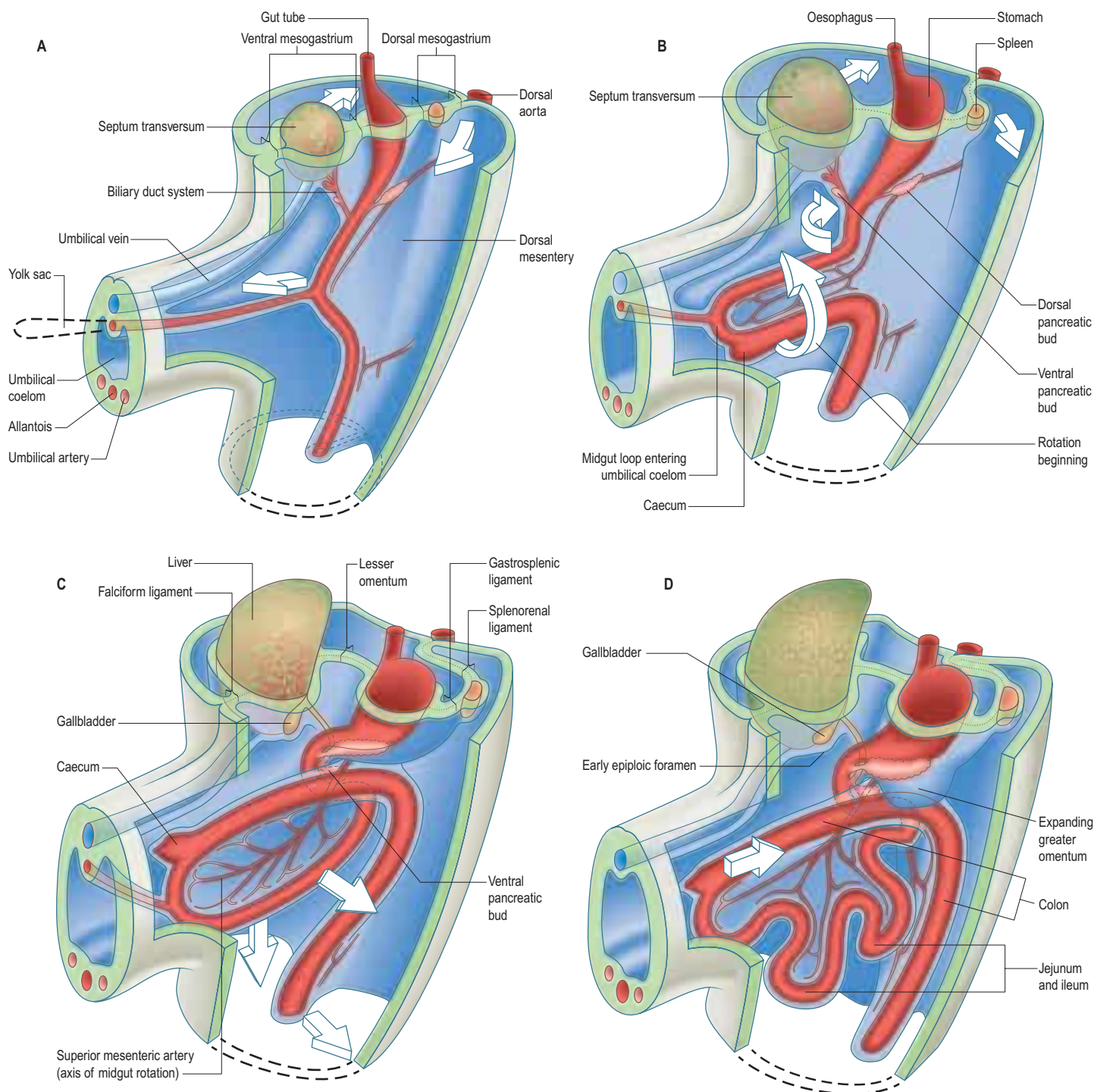


Fig. 60.6 The major developmental sequences of the subdiaphragmatic embryonic and fetal guts, together with their associated major glands, peritoneum and mesenteries: left anterolateral aspect. **A–F**, The development sequence spans 1.5 months to the perinatal period. **A–B**, The top white arrows show the relative movements of the dorsal and ventral mesogastria that result in the longitudinal rotation of the stomach and limited entry to the lesser sac (see also Fig. 60.5). **C–D**, The lower white arrows associated with the midgut indicate the relative movements and rotation of the midgut loop within the umbilical coelom, and as it returns to the abdominal cavity.

Continued

maturation. Developmental differences between parts of the small intestine or colon have not yet been correlated with age. The endodermal cells of the small intestine proliferate and form a layer some three to four cells thick, with mitotic figures throughout. From 7 weeks, blunt projections of the endoderm have begun to form in the duodenum and proximal jejunum; these are the developing villi, which increase in length until, in the duodenum, the lumen becomes difficult to discern. The concept of occlusion of the lumen and recanalization, which is described in many accounts of development, does not match the cyto-differentiation that occurs in the gut epithelia. Thus, it is no longer thought that there is secondary recanalization of the gut lumen. By 9 weeks, the duodenum, jejunum and proximal ileum have villi, and the remaining distal portion of ileum develops villi by 11 weeks. The villi are covered by a simple epithelium. Primitive crypts, epithelial downgrowths into the mesenchyme between the villi, appear between 10 and 12 weeks and similarly follow a craniocaudal progression. Brun-

ner's glands are present in the duodenum from 15 weeks and the muscularis mucosa can be seen in the small intestine from 18 weeks.

Whereas mitotic figures are initially seen throughout the endodermal layer of the small intestine prior to villus formation, by 10–12 weeks they are limited to the intervillous regions and the developing crypts. It is believed that an 'adult' turnover of cells may exist when rounded-up cells can be observed at the villus tips, in position for exfoliation. The absorptive enterocytes have microvilli at their apical borders before 9 weeks. An apical tubular system appears at this time, and is composed of deep invaginations of the apical plasma membrane and membrane-bound vesicles and tubules; many lysosomal elements (meconium corpuscles) appear in the apical cytoplasm. These latter features are more developed in the ileum than jejunum, are most prominent at 16 weeks, and diminish by 21 weeks. There are abundant deposits of glycogen in the fetal epithelial cells, and it has been suggested that, prior to the appearance of hepatic glycogen, the intestinal

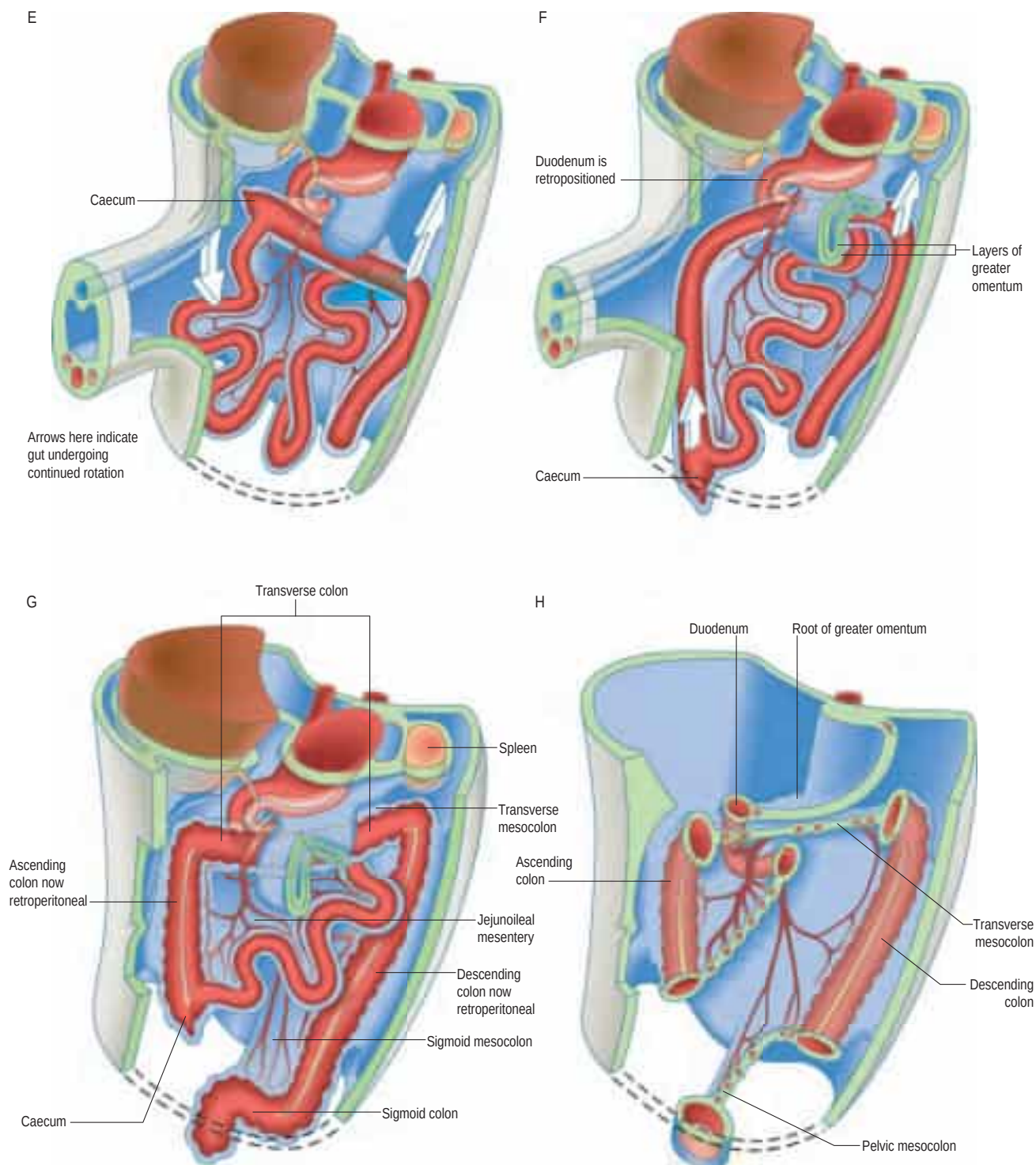


Fig. 60.6, cont'd The major developmental sequences of the subdiaphragmatic embryonic and fetal guts, together with their associated major glands, peritoneum and mesenteries: left anterolateral aspect. **E–F**, The lower white arrows associated with the midgut indicate the relative movements and rotation of the midgut loop within the umbilical coelom, and as it returns to the abdominal cavity. **G–H**, The approximate disposition in the adult abdomen of the gut (**G**) and the mesenteric roots, showing their lines of attachment and principal contained vessels (**H**).

epithelium serves as a major glycogen store. Goblet cells are present in small numbers by 8 weeks, Paneth's cells differentiate at the base of the crypts in weeks 11 and 12, and enteroendocrine cells appear between weeks 9 and 11. M cells (membrane or microfold cells) are present from 14 weeks.

Intestinal subepithelial myofibroblasts (ISEMFs) have been described within the villus cores and adjacent to the intervillous space at the base of the crypts from week 21. They are arranged as a syncytium between the epithelium and the muscularis mucosae, where they contribute to the extracellular matrix. They demonstrate α -smooth muscle actin (SMA) expression (McLin et al 2009). It is not clear whether these cells derive from splanchnopleuric mesenchyme fibroblasts, smooth muscle precursors or neural crest. They migrate from the crypts to the villous axis in a manner similar to enterocytes.

Meconium can be detected in the lumen of the intestine by the sixteenth week. It is derived from swallowed amniotic fluid, which

contains vernix and cellular debris, salivary, biliary, pancreatic and intestinal secretions, and sloughed enterocytes. As the mixture passes along the gut, water and solutes are removed and cellular debris and proteins concentrated. Meconium contains enzymes from the pancreas and proximal intestine in higher concentrations in preterm than in full-term babies.

Muscularis layer

The muscularis layer is derived from the splanchnopleuric mesenchyme, as it is in other parts of the gut. Smooth muscle myoblasts are characterized by expression of α -SMA. Longitudinal muscle can be seen from 12 weeks. At 26–30 weeks, the gut shows contractions without regular periodicity; from 30–33 weeks, repetitive groups of regular contractions have been seen in preterm neonates.

Serosa

The small intestine possesses only a dorsal mesentery. The movement of the root of this dorsal mesentery, and the massive lengthening of its enteric border in order to match the longitudinal growth of the gut tube, reflect the spiralizing of the midgut loop in the umbilical coelom.

Small intestine at birth

The radial patterning of the small intestine is completed before birth, with differentiation of the crypt–villous axis. Specification of the space between villi, crypt depth and villous length is a dynamic process dependent on the establishment of the intestinal microbiota. In the neonate, the small intestine forms an oval-shaped mass with its greater diameter transversely orientated in the abdomen, rather than vertically as in the adult. The mass of the small intestine inferior to the umbilicus is compressed by the urinary bladder, which is anterior at this point. The small intestine is 300–350 cm long at birth and its width, when empty, is 1–1.5 cm. The ratio between the length of the small and the length of the large intestine at birth is similar to the adult ratio. The mucosa and submucosa are fairly well developed and villi are present throughout the small intestine; however, some epithelial differentiation is incomplete. The muscularis is very thin, particularly the longitudinal layer, and there is little elastic tissue in the wall. There are few or no circular folds in the small intestine, and the jejunum and ileum have little fat in their mesentery.

PRIMITIVE HINDGUT

Just as the foregut has an extensive ventral endodermal diverticulum, which contributes to a system separate from the gut, so, too, the hindgut has a ventral diverticulum – the allantois – destined for a different system. However, unlike the respiratory diverticulum of the foregut, the allantois is formed very early in development, prior even to formation of the embryonic endoderm and tail-folding. With the reorganization of the caudal region of the embryo at stage 10, part of the allantois is drawn into the body cavity. The early embryonic hindgut thus consists of a dorsal tubular region extending from the caudal intestinal portal to the cloacal membrane, and a ventral blind-ending allantois extending from the cloacal region into the connecting stalk. The slightly dilated cavity, lined by endoderm, that cranially receives the enteric hindgut proper and the root of the allantoenteric diverticulum is termed the endodermal cloaca. It is closed ventrally by the cloacal membrane (endoderm opposed to proctodeal ectoderm), and it also has, transiently, a small recess of endoderm in the root of the tail, the postanal gut. As elsewhere, the hindgut, allantois and endodermal cloaca are encased in splanchnopleuric mesenchyme. Proliferation of the mesenchyme and endoderm in the angle of the junction of hindgut and allantois produces a urorectal septum (see Fig. 72.4B). Continued proliferation of the urorectal septum and elongation of the endodermal structures thrust the endodermal epithelium towards the cloacal membrane, with which it fuses centrally, separating the presumptive rectum and upper anal canal (dorsally) from the presumptive urinary bladder and urogenital sinus (ventrally) (see Fig. 72.4C). The cloacal membrane is thus divided into anal (dorsal) and urogenital (ventral) membranes. The nodal centre of division is the site of the future perineal body, the functional centre of the perineum.

ENTERIC HINDGUT

The development of the large intestine, whether derived from mid- or hindgut, seems to be similar. The proximal end of the colon can be first identified at stage 15, when an enlargement of a local portion of gut on the caudal limb of the midgut loop defines the developing caecum. An evagination of the distal portion of the caecum forms the vermiform appendix at stage 17 (see Fig. 60.3E). Apart from the embryonic studies of Streeter (1942), there is little information about the development of the large intestine in humans. The early endodermal lining of the colon appears stratified, and mitoses occur throughout the layers. A series of longitudinal folds arise initially at the rectum and caecum, and later in the regions of colon between these two points. The folds segment into villi with new villi forming between. The developing mucosa invaginates into the underlying mesenchyme between the villi to form glands that increase in number by splitting longitudinally from the base upwards. The villi gradually diminish in size and are absent by the time of birth.

MATURATION OF THE LARGE INTESTINE

The similarity of development of the small and large intestines is further mirrored in their cytological differentiation. Fetal gut, from 11 weeks, shows dipeptidase activity in the colon as well as in the small intestine. Throughout preterm development, meconium corpuscles are seen in the colon and in the small intestine; they are believed to be the phagocytosed remains of neighbouring cells that have died as a result of programmed cell death.

There is little direct evidence of colonic function in the human fetus and neonate. However, the specific results of mammalian studies are being correlated to human studies where possible. A number of distinct and important differences between the function of adult and fetal colon have been reported.

Mucosa

The absorption of glucose and amino acids does not take place through the colonic mucosa in adult life, but there is evidence of direct absorption of these nutrients during development. At birth, the normal cycle of bile acids is not mature. In the adult, bile is secreted by the liver, stored in the gallbladder and then secreted into the intestine, where it is absorbed by the jejunum and ileum. In the fetus and neonate, the transport of bile acids by an active process from the ileum does not occur, and so bile salts pass on into the colon. In the adult, the presence of bile salts in the colon stimulates the secretion of water and electrolytes, which results in diarrhoeal syndrome; however, the fetal and neonatal colon seems protected from this effect. The colon is not considered a site of significant nutrient absorption in the adult, and yet neonates are unable to assimilate the full lactose load of a normal breast feed from the small intestine and a large proportion of it may be absorbed from the colon. Thus, it appears that the colon fulfils a slightly different role in the preterm and neonatal period, conserving nutrient absorption and minimizing fluid loss until the neonate has adjusted to extrauterine life, oral feeding and the establishment of the symbiotic bacterial flora.

Muscularis

The muscularis is present and functioning by the eighth week, when peristaltic waves have been observed. The specific orientation of the longitudinal muscle layer into taeniae coli occurs in the eleventh to twelfth weeks, when haustra appear. The enteric nerves are present in Meissner's and Auerbach's plexuses at 8 and 12 weeks, respectively; there is a craniocaudal migration of neurones into the gut wall. A normal distribution of ganglion cells has been noted in preterm babies of 24 weeks, although there is a region devoid of ganglia just above the anal valves. Anomalous migration of neural crest cells to the gut may give rise to Hirschsprung's disease (see below). Puborectalis appears in 20–30 mm embryos, following opening of the anal membrane.

Serosa

The development of the serosa of the intestine is considered with the development of the peritoneal cavity (see below).

Colon at birth

In the neonate, the colon is typically 66 cm long and averages 1 cm in width. The caecum is relatively smaller than in the adult; it tapers into the vermiform appendix. The ascending colon is shorter in the neonate, reflecting the shorter lumbar region. The transverse colon is relatively long, whereas the descending colon is short, but twice the length of the ascending colon (see Fig. 14.6B). The sigmoid colon may be as long as the transverse colon; it often touches the inferior part of the anterior body wall on the left and, in about half of neonates, part of the sigmoid colon lies in the right iliac fossa. The muscularis, including the taeniae coli, is poorly developed in the colon, as it is in the small intestine. Appendices, epiploicae and haustra are not present, which gives a smooth external appearance to the colon. Haustra appear within the first 6 months. The rectum is relatively long; its junction with the anal canal forms at nearly a right angle.

ANAL CANAL

Mesenchymal proliferation occurs around the rim of the ectodermal aspect of the anal membrane, which thus comes to lie at the bottom

of a depression, the proctodeum (see Fig. 72.7E). With the absorption and disappearance of the anal membrane, the anorectum communicates with the exterior. The lower part of the anal canal is formed from the proctodeal ectoderm and underlying mesenchyme, but its upper part is lined by endoderm. The line of union corresponds with the edges of the anal valves in the adult. The dual origin of the anal canal is reflected in its innervation: the endodermal portion is innervated by autonomic nerves, and the ectodermal proctodeum is innervated by spinal nerves.

In the fourth and fifth weeks, a small part of the hindgut, the post-anal gut, projects caudally beyond the anal membrane (see Fig. 60.3B); it usually disappears before the end of the fifth week.

Imperforate anus is a term used to describe many different anorectal anomalies. The most common is anal agenesis, which is found in almost half of all cases of imperforate anus. The condition is usually associated with a fistula, which opens into the vulva (females) or into the urethra (males). It is more rare for the anal membrane to fail to perforate. The condition cannot reliably be diagnosed prenatally by ultrasound diagnosis, and it may be confused with Hirschsprung's disease (see below) and colonic atresia. The prognosis is good for low lesions of the anal canal. The principal concern, in all cases, is the degree of bowel control, urinary control and, in some cases, sexual function, which is compromised by the condition. Anorectal anomalies may be indicators of other anomalies, e.g. those forming the 'VATER' syndrome (vertebral, anal, tracheo-oesophageal and renal anomalies).

ENTERIC NERVOUS SYSTEM

Enteric neurones are derived from cranial (vagal) neural crest cells at somite levels 1–7 and from 28 onwards (see Fig. 17.19 and p. 251). After neurulation, the crest cells begin their ventral migration and invade the gut via the dorsal mesentery (see Fig. 17.11). Glial cells associated with the gut have been identified as arising from similar levels. It is thought that the vagal and sacral neural crest cells have intrinsic differences in their ability to colonize the gut. The local splanchnopleuric mesenchyme patterns the crest cells, such that those that enter the gut layers attain an enteric fate, whereas those that remain outside the gut become committed as parasympathetic postganglionic neurones. The enteric neurones also migrate to the glands of the gut, e.g. the pancreas. Migrating vagal neural crest cells reach the midgut by week 5 and the entire length of the gut is colonized by week 7. Maturation of the myenteric plexus follows a craniocaudal progression. The submucosal plexus arises from centripetally migrating cells from the myenteric plexus 2–3 days later (Burns et al 2009). Interstitial cells of Cajal, which derive from splanchnopleuric mesenchyme, do not mature in a craniocaudal progression. They are seen around the myenteric ganglia along the entire gut from weeks 12–14 (Burns et al 2009). There is evidence that mucosal enteric glial cells, which move into the mucosa after birth, interact with the colonizing intestinal flora, suggesting a microbiota-driven homeostatic mechanism for gut function (Kabouridis et al 2015).

Hirschsprung's disease

Hirschsprung's disease (commonly called megacolon) is usually characterized by an aganglionic portion of gut that does not display peristalsis, and a dilated segment of structurally normal colon proximal to this site. Histologically, there is either an absence or a reduction in the number of ganglia and postganglionic neurones in the myenteric plexus of the affected segment of gut; postganglionic innervation of the muscle layers is also often defective. It is believed that the condition is caused by a failure of neural crest cells to colonize the gut wall appropriately (Gershon 2010). A variable length of large intestine may be affected; the lower and mid-rectum are the most common sites but, in severe cases, the rectum, sigmoid, descending and even proximal colon can be aganglionic. Occasionally, aganglionosis affects only a very short length of rectum proximal to the anorectal junction and the degree of functional obstruction is minimal; in these cases of 'ultra-short-segment Hirschsprung's disease', clinical anomalies arise later in life. Infants with Hirschsprung's disease show delay in the passage of meconium, constipation, vomiting and abdominal distension.

GUT-ASSOCIATED LYMPHOID TISSUE

Individual lymphocytes appear in the lamina propria of the gut from approximately week 12 of development, and lymphoid aggregates – Peyer's patches – have been noted between 15 and 20 weeks; it is not

clear whether these cells migrate in from distant sources or differentiate from the investing mesenchyme. The endodermal epithelium overlying the lymphoid aggregates is often distorted into a dome shape. The cells within the dome are a mixed population of enterocytes, endocrine cells, goblet cells and M cells. M cells are specialized to provide a mechanism for the transport of microorganisms and intact macromolecules across the epithelium to the intraepithelial space and lamina propria. They have been observed in the fetus by 17 weeks; it is believed that they are formed by a specialized epithelial–mesenchymal interaction of the endoderm and underlying lymphoid-type mesenchyme.

There are similarly specialized epithelial cells between the enterocytes. Intraepithelial leukocytes typically account for 15% of the epithelial cells of the gut in the adult. They have been observed at 11 weeks, with a distribution of 2–3 intraepithelial leukocytes per 100 gut epithelial cells. Both T and B lymphocytes have been described in the developing gut wall. For an account of the development of the immune cells of the gut, consult Butzner and Befus (1989). The neonatal gut becomes colonized by a range of bacterial flora; some of these exist in a symbiotic relationship with their host, and some of them may be considered pathogenic.

ULTRASOUND ANTENATAL IMAGING OF THE FETAL GUT

The abdominal circumference, routinely estimated at 20 weeks, is calculated from the anteroposterior diameter and the transverse abdominal diameter perpendicular to it, taken at the widest part of the abdomen. The section usually includes one entire rib and the stomach cavity (termed stomach bubble) and can be confirmed on the left side of the body (see Fig 14.4G). Anteriorly, the connection of the umbilical cord can be confirmed (see Fig 14.4H). Anomalies of the anterior body wall, exomphalos and gastroschisis, are readily detected. In exomphalos, a mass of peritoneal organs herniate through the base of the umbilical ring; in gastroschisis, loops of small intestine and colon float free in the amniotic fluid (see Fig. 14.5B) (Ionescu 2014).

FUNCTIONAL MATURITY OF THE GUT AT BIRTH

The postnatal maturation of the gut is dependent on the establishment of an intestinal microbiota, which, in turn, is affected by gestational age, mode of delivery, type of feeding and any medical interventions. In cases of chorioamnionitis, preterm infants ingest bacterial products from the amniotic fluid; it has been suggested that this may be associated with preterm labour (Neu and Mai 2012). Normal, term, vaginal delivery establishes the initial colonization of the neonatal gut with maternal vaginal and intestinal flora. Two-way interactions between enteric microbes in a biofilm within the luminal glycocalyx of enterocytes modify intestinal permeability, increase T and B lymphocyte numbers within the mucosal lamina propria and mesenteric lymph nodes, and stimulate postnatal immune development (Vaarala 2012, Wynn and Neu 2012, Martin et al 2012, Patel et al 2012, Grütz and Bhandari 2015). Delivery by caesarean section results in the establishment of skin microbiota, different intestinal microbes and low microbiota diversity. Failure to develop an appropriate, dynamic, microbiota is thought to be associated with allergic and inflammatory conditions in later life (Wynn and Neu 2012). (For further reading, see Neu (2012), Bäckhed et al (2015), Rodríguez et al (2015).)

The onset of feeding also contributes to gut maturity. Breast milk is a source of epidermal growth factor, which promotes postnatal mucosal development and helps mucosal repair. Breast feeding also affects the production of immunoglobulin A (IgA) in the gut mucosa (Vaarala 2012). Undernutrition leads to villous atrophy and, in the preterm infant, parenteral nutrition is associated with changes in intestinal structure and function that may lead to increased intestinal permeability (Martin et al 2012). Antibiotic therapy within the neonatal period can lead to changes in, and the establishment of, the normal intestinal microbiota (Martin et al 2012).

The first passage of faeces of the newborn is termed meconium. This is a dark, sticky, viscid substance formed from the passage of amniotic fluid, sloughed mucosal cells, digestive enzymes and bile salts along the fetal gut. Meconium becomes increasingly solid as gestation advances but does not usually pass out of the fetal body while *in utero*. Fetal distress produced by anoxia may induce the premature defecation of meconium into the amniotic fluid, with the risk of its inhalation. At birth, the colon contains 60–200 g of meconium. The majority of neonates defecate within the first 24 hours after birth. Delayed passage of stool beyond this time is associated with Hirschsprung's disease (see

above) or imperforate anus. The normal passage of meconium continues for the first 2 or 3 days after birth, and is followed by a transition to faecal stools by day 7.

PERITONEAL CAVITY

The early development of the intraembryonic coelom, which gives rise to the peritoneal cavity, is described on [page 185](#). [Figure 12.2](#) shows a scheme of the shape of the early peritoneal cavity and indicates the mesenchymal populations derived from its epithelial walls. Initially, the peritoneal cavity associated with the lower end of the foregut has separate right and left components: the pleuroperitoneal canals (see [Fig. 60.1](#)). At the level of the midgut, the pleuroperitoneal canals join a confluent cavity surrounding the developing gut, which transitorily is in communication with the extraembryonic coelom.

The description of the development of the peritoneal cavity that follows pertains to changes that occur as a consequence of the differential growth of the gut.

PERITONEUM

Peritoneum develops from a specific portion of the intraembryonic coelomic walls. Initially, the intraembryonic coelomic epithelium is a pseudostratified germinal layer from which cellular progeny with different fates arise in specific sites and at specific developmental times. The portion that will give rise to the peritoneum is derived from the lower portion of the pericardioperitoneal canals and the somatopleure and splanchnopleure associated with the lower foregut, midgut and upper portions of the hindgut (see [Figs 12.2, 60.1](#)).

The proliferative splanchnopleuric epithelium produces cell populations for the mucosa and muscularis of the gut, and also for the lamina propria and epithelium of the visceral peritoneum (the serosa of the gut wall). The somatopleuric epithelium gives rise to the lamina propria and epithelium of the parietal peritoneum. The visceral and parietal peritoneal layers constitute a mesothelium, which denotes their origin from the intraembryonic mesoderm of the coelomic wall.

As the gut grows, splanchnic mesenchyme accumulates around the endodermal epithelium and the whole unit generally moves ventrally. There is a concomitant enlargement of the caudal ends of the developing pericardioperitoneal (pleuroperitoneal) canals and developing peritoneal cavity. The medial walls of the intraembryonic coelom move closer and there is a relative decrease in the mesenchyme between them. The regions where the medial portions of the intraembryonic coelom come together are termed mesenteries. They are composed of two layers of peritoneum with intervening mesenchyme and contain the neurovascular structures that pass to and from the gut. At the caudal ends of the pleuroperitoneal canals, the gut has both ventral and dorsal mesenteries, whereas, caudal to this, there is only a dorsal mesentery (see [Fig. 60.6](#); [Fig. 60.7](#)).

The mesenteries attached to the gut lengthen to permit large movements or rotations of the gut tube. Later, when part or the whole of the mesentery lies against the parietal peritoneum, their apposed surfaces fuse and are absorbed, i.e. they become sessile. Only those viscera developed in direct apposition to one of the primary coelomic regions, or a secondary extension of the latter, retain a partial or almost complete visceral serous cover. Thus, the original line of reflection of mesenteries becomes altered, or, in some cases, the organ may become retroperitoneal. These mechanisms are significant throughout the subdiaphragmatic gut, but are predominant in the small and large intestine. All serous membranes may vary their thickness, lines of reflection, disposition, 'space' enclosed and their channels of communication, by a process of areal and thickness growth on one aspect, combined with cavitation (leading to expanding embryonic recess formation) on the other.

Although all of the gut tube and its derived glands are associated with mesenteries formed as described above, the nomenclature for some portions of the gut and glands is different. Thus, the mesenteries of the stomach are called omenta and the reflections of peritoneum around the liver, which develop from a confluence of splanchnopleuric, somatopleuric and septum transversum-derived portions, are called ligaments.

The movements of the developing viscera within the peritoneal cavity occur with associated movements of the mesothelia that surround them. The descriptions of peritoneal cavity development that follow are thus describing a sequence of changes that affect a complex space and its boundaries.

Mesenteries of the developing gut

The cervicothoracic oesophagus develops between the pericardioperitoneal canals. It is encased in prevertebral, retrotracheal and retrocardiac mesenchyme. As the pericardioperitoneal canals expand with the developing lung buds, and the diaphragm forms immediately below them; the oesophagus, at this level, has no true dorsal or ventral mesentery. At superior and intermediate thoracic levels, parts of the lateral aspects of the oesophagus are closely related to the secondary, mediastinal, parietal pleura. In the lower thorax, the oesophagus inclines ventrally anterior to the descending thoracic aorta. The dorsocaudally sloping midline diaphragm between oesophageal and aortic orifices may be homologized with part of a dorsal meso-oesophagus, and is used in that context in descriptions of diaphragmatic development. A ventral midline diaphragmatic strip may also be considered to be a derivative of a ventral meso-oesophagus; however, this region is more usually thought of as septum transversum.

The alimentary tube, from the diaphragm to the start of the rectum, initially possesses a sagittal dorsal mesentery. Its line of continuity with the dorsal parietal peritoneum (i.e. its 'root' or 'line of reflection') is, initially, also midline.

The abdominal foregut, from the diaphragm to the future hepatopancreatic duodenal papilla, also has a ventral mesentery. This extends from the ventrolateral margins of the abdominal oesophagus and as yet 'unrotated' primitive stomach and proximal duodenum, cranially to the future diaphragm and anteriorly to the ventral abdominal wall (to the level of the cranial rim of the umbilicus). Caudally, between umbilicus and duodenum, it presents a crescentic free border.

The midgut and hindgut have no ventral mesentery; thus, the pleural and supra-umbilical peritoneal cavities are, initially and transiently, bilaterally symmetrical above the umbilicus. Below the umbilicus, the peritoneal cavity is freely continuous across the midline ventral to the gut (see [Fig. 60.5A](#)).

Foregut mesenteries

The ventral and dorsal foregut mesenteries are relatively large compared with the slender endodermal tubes they encase; they are composed of mesenchyme sandwiched between two layers of splanchnopleuric coelomic epithelium. A complex series of recesses develop in the splanchnopleuric mesenchyme and become confluent. As a result of foregut rotation, differential growth of the stomach, liver, pancreas and spleen, and the completion of the diaphragm, the territories of the greater sac and lesser sac (omental bursa) are delimited, and the mesenteric complexes of these organs (omenta and 'ligaments') are defined (see [Figs 60.6–60.7](#)).

Consequences of rotation of the stomach

A number of processes occur concurrently, which, conceptually, can be visualized as the movement of the right pleuroperitoneal canal to a position posterior to the stomach, such that its communication with the remainder of the peritoneal cavity is reduced. These processes include the differential growth of the walls of the stomach, the formation and specific local extension of the omenta (dorsal and ventral mesogastria), and the growth of the liver and, particularly, of the vessels and ducts that enter and leave the liver. These developments permit stomach expansion both anteriorly and posteriorly when food is ingested, and free movement of peristalsis. The right pleuroperitoneal canal forms a discrete region of the peritoneal cavity, the lesser sac, and the remaining left pleuroperitoneal canal and the remainder of the peritoneal cavity form the greater sac. The entrance to the original right pleuroperitoneal canal (lesser sac) becomes reduced in size. It is called the epiploic foramen, foramen of Winslow, or the aditus of the omental bursa (bursa omentalis).

Early stages of lesser sac development

The lesser sac is first indicated by the appearance of multiple clefts in the para-oesophageal mesenchyme on both left and right aspects of the oesophagus. Although they may become confluent, the left clefts are transitory and soon atrophy. The right clefts merge to form the right pneumatoenteric recess that extends from the oesophageal end of the lesser curvature of the stomach as far as the caudal aspect of the right lung bud. At its gastric end, it communicates with the general peritoneal cavity and lies ventrolateral to the gut; more rostrally it lies directly lateral to the oesophagus. It is not, as commonly stated, a simple progressive excavation of the right side of the dorsal

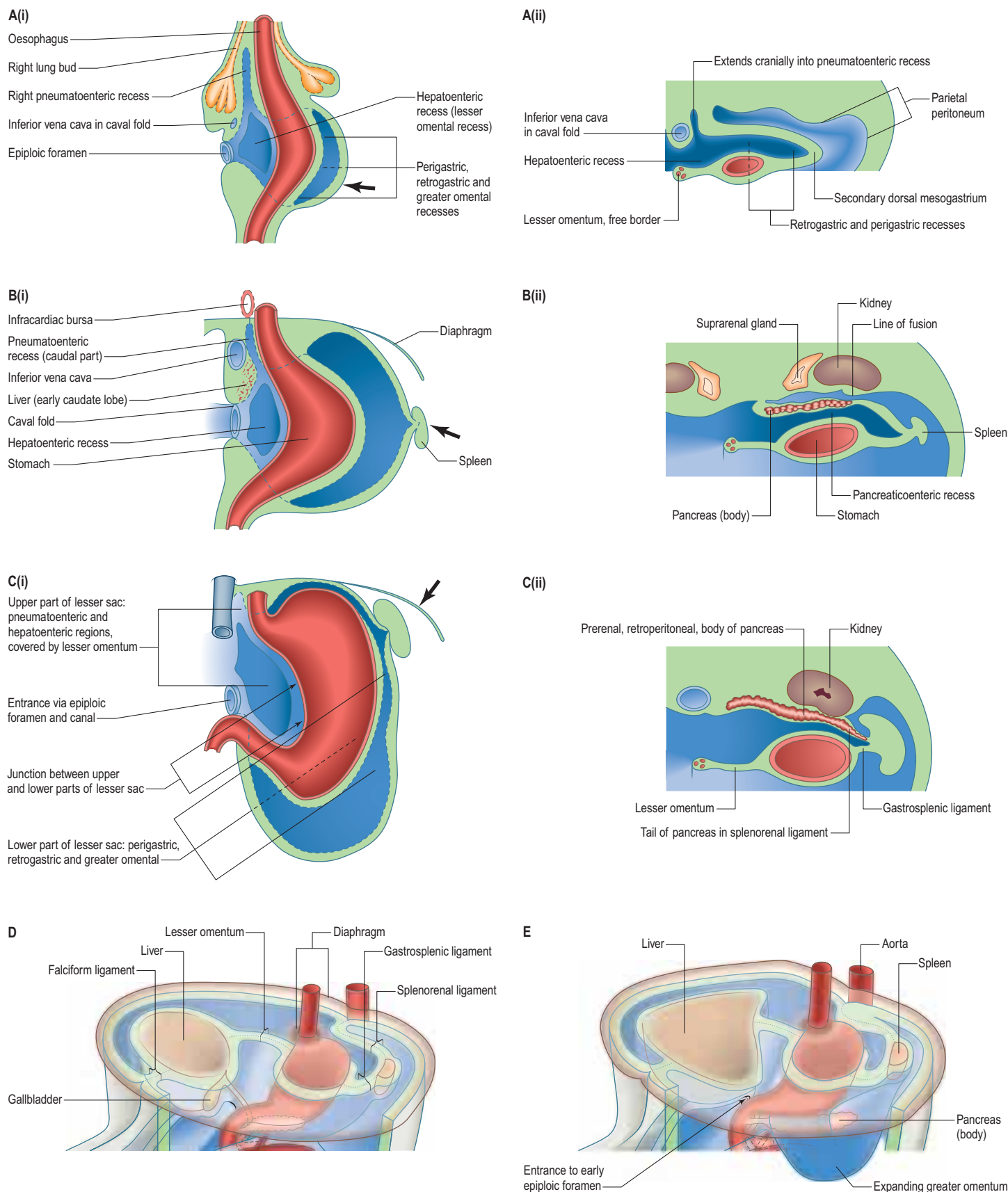


Fig. 60.7 A–C, Stages of development of the subdiaphragmatic foregut and the right and left pericardioperitoneal/pleuroperitoneal canals, with particular reference to the terminal oesophagus, stomach, duodenum, spleen, the lesser sac of the peritoneum and the omenta, seen in semicoronal section (left column) and transverse section at the levels indicated by the arrows (right column). **D–E,** The lesser sac and dorsal and ventral mesogastrica.

mesogastrium. The right pneumatoenteric recess undergoes further extension, subdivision and modification (see Fig. 60.7 A(i), B(i)). From its caudal end, a second process of cleft and cavity formation occurs, which produces the hepatoenteric recess. This thins and expands the splanchnopleure between the liver and the stomach and proximal duodenum, and reaches the diaphragm (see Fig. 60.7A(i), C(i)). The resulting, structurally bilaminar, mesenteric sheet is the lesser omentum. It is derived, cranially to caudally, from the small

meso-oesophagus; the much larger ventral mesogastrium, and the most caudal free border, is from the ventral mesoduodenum. As differential growth of the duodenum occurs, the biliary duct is repositioned and most of the duodenum becomes sessile. The duodenal attachment of the free border and a continuous neighbouring strip of the lesser omentum become confined to the upper border of a short segment of its superior part. The contrasting growth and positioning of its attached viscera cause the free border to change gradually from the

horizontal to the vertical. It carries the bile duct, portal vein and hepatic artery, and its hepatic end is reflected around the porta hepatis. An alternative name for this part of the lesser omentum is the hepatoduodenal ligament; it forms the anterior wall of the epiploic foramen. The floor of the foramen is the initial segment of the superior part of the duodenum, its posterior wall is the peritoneum covering the immediately subhepatic part of the inferior vena cava, and its roof the peritonealized caudate process of the liver. The major part of the lesser omentum from the lesser gastric curvature passes in an approximately coronal plane to reach the floor of the increasingly deep groove for the ductus venosus on the hepatic dorsum; this part is sometimes called the hepatogastric ligament.

Ligaments of the liver

The liver is precociously large during development because of its early role in haemopoiesis. Thus, the liver mass projects into the abdominal cavity with equal growth on the two sides of the peritoneal cavity. The ligaments associated with the liver develop from the ventral mesogastrium – which passes from the foregut to the ventral abdominal wall, down to the cranial rim of the intestinal portal – and from the reflections of peritoneum from the liver to the diaphragm.

The medial portions of the germinative coelomic epithelial walls, containing splanchnopleuric mesenchyme, septum transversum mesenchyme and developing liver, constitute the early ventral mesogastrium (see Fig. 60.5). The mesenchyme between these layers is continuous superiorly with the septum transversum mesenchyme of the diaphragm. The coelomic epithelial layers of the ventral mesogastrium almost touch anterior and posterior to the liver, and are separated by a slender lamina of mesenchyme. They form the falciform ligament and the lesser omentum, respectively, and where they are in contact with the liver directly, they form visceral peritoneum (see Fig. 60.7D).

When the diaphragm is formed above the liver, local cavities coalesce and open into the general coelomic cavity as extensions of the greater (and lesser) sacs. In this way, almost all the ventrosuperior, visceral and some of the posterior aspects of the liver become peritonealized. The process of extending the greater sac continues over the right lobe and stops when the future superior and inferior layers of the coronary ligament and the right triangular ligament are defined. Those, plus a medial boundary provided by an extension of the lesser sac, enclose the 'bare area' of the liver, where loose areolar tissue of septum transversum origin persists. Later in development, when the haemopoietic function of the liver declines, the left lobe becomes relatively smaller than the right; this, presumably, accounts for the smaller size of the left triangular ligament.

Where the superior layers of the coronary and left triangular ligaments meet, they continue as a (bilaminar) ventral mesentery attached to the ventrosuperior aspects of the liver. Its somewhat arched umbilico-hepatic free caudal border carries the left umbilical vein. As the ventral body wall develops, this falciform ligament, which initially attaches to the early cranial intestinal portal, is drawn to the diminishing cranial rim of the umbilicus. It may be considered the final ventral part of the ventral mesogastrium, although its free border has a ventral mesoduodenal origin. Its passage to the ventral body wall becomes increasingly oblique, curved and falciform (sickle-shaped) as the umbilicus becomes more defined.

In the early embryo, the connection between one pericardioperitoneal canal and the other was directly across the ventral surface of the cranial midgut, immediately caudal to the developing primitive ventral mesogastrium. By stage 14, the passage from one side of the falciform ligament to the other necessitates passing below the greatly enlarged liver, or the curved lower edge of the falciform ligament, or the lesser omentum. The position of the falciform ligament is of clinical interest in the neonate in diagnosing pneumoperitoneum because it is silhouetted by air on abdominal X-rays.

Caval fold

The caval fold is a linear eminence, with divergent cranial and caudal ends, which passes from the upper abdominal to the lower thoracic region and protrudes from the dorsal wall of the pleuroperitoneal canal. Cranially, it becomes continuous, lateromedially, with the root of the pulmonary anlage and pleural coelom, the uppermost portion of the septum transversum mesenchyme, and the retrocardiac mediastinal mesenchyme. Caudally, it forms an arch with dorsal and ventral horns. The dorsal horn merges with the primitive dorsal mesentery and the mesonephric ridge and associated gonad and suprarenal (adrenal) gland. The ventral horn is confluent with the dorsal surface of the septal mesenchyme.

Thus, the caval fold is a zone where intestinal, mesenteric, intermedial, hepatic, pericardial, pulmonary and mediastinal mesenchymes

meet and blend. It provides a mesenchymal route for the upper abdominal, transdiaphragmatic and transpericardial parts of the inferior vena cava, and it is also prominent in the development of parts of the liver, lesser sac of peritoneum, and certain mesenteries. The left fold regresses whereas the right fold enlarges rapidly (see Fig. 60.7A).

The pneumatoenteric recess continues to expand to the right into the substance of the caval fold. It stops near the left margin of the hepatic part of the inferior vena cava, which remains extraperitoneal and crosses the base of the now roughly triangular bare area of the liver and this new expanded line of reflection. With closure of the pleuro-peritoneal canals, the rostral part of the right pneumatoenteric recess is sequestered by the diaphragm but often persists as a small serous sac in the right pulmonary ligament. The remaining caval fold mesenchyme to the left of the inferior vena cava – which forms the right wall of the upper part of the lesser sac – becomes completely invaded by embryonic hepatic tissue and is transformed into the caudate lobe of the liver. This smooth, vertically elongated, mass projects into the cavity of the lesser sac; both its posterior, and much of its anterior, surfaces become peritonealized as a result of the increasing depth of the groove for the ductus venosus and the attachment of the lesser omentum to its floor.

Later stages of lesser sac development

The lower (inferior) part of the lesser sac can be first seen in embryos of 8–9 mm crown–rump length, and the early pneumatoenteric and hepatoenteric recesses are well established. Progressive differential gastric growth produces an elliptical transverse sectional profile, with a right-sided lesser curvature, which corresponds to the original ventral border of the gastric tube. The lesser omental gastric part of the ventral mesogastrium remains attached to this border. The greater curvature of the stomach is a new, rapidly expanding, region; its convex profile projects mainly to the left, but also cranially and caudally (see Fig. 60.7A(i), B(i), C(ii)). The original dorsal border of the gastric tube now traverses the dorsal aspect of the expanding rudiment, curving along a line near the lesser curvature. The primitive dorsal mesogastrium is transiently attached to it, and blends with the thick layer of compound gastric mesenchyme clothing the posterior aspect and greater curvature of the miniature stomach. Because of its thickness, the mesenchyme projects cranially, caudally and, particularly, to the left, beyond the 'new' greater curvature of the endodermal lining of the stomach.

The processes already described in relation to the ventral mesenteries now supervene. Multiple clefts appear at various loci in the mesenchyme, and there are local mesenchyme to epithelial transitions. The groups of clefts rapidly coalesce to form transiently isolated closed spaces, which soon join with each other and with the preformed upper part of the lesser sac; the newly formed epithelia join the coelomic epithelium. In sequence, the initial loci occur in the compound posterior gastric mesenchyme nearer the lesser curvature and along its zone of blending with the primitive dorsal mesogastrium; in the dorsal mesoduodenum; and independently, in the caudal rim, where greater curvature mesenchyme and dorsal mesogastrium blend. As these cavities become confluent and their 'reniform' expansion follows, matches and then exceeds that of the gastric greater curvature, there are several major sequelae. The primitive dorsal mesogastrium increases in area by intrinsic growth, and, as cavitation proceeds, by incorporating substantial contributions from the dorsal lamella separated by cleavage of the posterior gastric mesenchyme; it is now called the secondary dorsal mesogastrium (see Fig. 60.7A(ii)). The gastric attachment of the secondary dorsal mesogastrium changes progressively. It may be regarded as a set of somewhat spiral lines, longitudinally disposed, that move with time to the left, from near the lesser curvature, towards and finally reaching the definitive greater curvature. The parietal mesogastric and (cleaving) mesoduodenal attachment remains in the dorsal midline for a time, but it later undergoes profound changes. With the confluence of the cavities that collectively form the lower part of the lesser sac, its communication with the upper part (which corresponds to the lesser gastric curvature and right and left gastropancreatic folds) becomes better defined. Ventral to the lower part of the cavity, post-cleavage splanchnopleure covers the posteroinferior surface of the stomach and a short proximal segment of the duodenum. This ventral wall is continued beyond the greater curvature and duodenum as the splanchnopleuric strip of visceral attachment of the secondary dorsal mesogastrium and mesoduodenum. The radial width of the strip is relatively short cranially (gastric fundus) and gradually increases along the descending left part of the greater curvature. It is longest throughout the remaining perimeter of the greater curvature as far as the duodenum; this prominent part shows continued marginal (caudoventral and lateral) growth

with extended internal cavitation (its walls constitute the expanding greater omentum; see Fig. 60.6H). The margins of the cavity of the inferior part of the lesser sac are limited by the reflexed edges of the ventrally placed strata derived from the secondary dorsal mesogastrium just described. These converge to form the splanchnopleuric dorsal wall, which is initially 'free' throughout, except at its midline dorsal root. At roughly mid-gastric levels, the pancreatic rudiment grows obliquely encased in this dorsal wall; its tail ultimately reaches the left limit of the lesser sac at the level of the junction between gastric fundus and body (see Fig. 60.7C(ii)).

Greater omentum

The greater omentum continues to grow, both laterally and, particularly, caudoventrally. It covers and is closely applied to the transverse mesocolon, transverse colon and inframesocolic and infracolic coils of small intestine (see Fig. 60.6D–G). At this stage, the quadrilaminar nature of the dependent part of the greater omentum is most easily appreciated. 'Simple' mesenteries are bilaminar: they possess two mesothelial surfaces derived from splanchnopleuric coelomic epithelium, which enclose a connective tissue core derived from splanchnopleuric mesenchyme. In the greater omentum, the gastric serosa covering its postero-inferior surface (single mesothelium) and the anterosuperior serosa (single mesothelium) converge to meet at the greater curvature and initial segment of the duodenum. The resulting bilaminar mesentery continues caudoventrally as the 'descending' stratum of the omentum. This, on reaching the omental margins, is reflexed and now passes craniodorsally to its parietal root as the 'ascending' posterior bilaminar stratum. The two bilaminar strata are, initially, in fairly close contact caudally, but are separated by a fine, fluid-containing, cleft-like extension of the lower part of the lesser sac. The posterior mesothelium of the posterior stratum makes equally close contact with the anterosuperior surface of the transverse colon, starting at the taenia omentalis, and with its transverse mesocolon.

Maturation of the lesser sac

At this stage, and subsequently, it is convenient to designate the lower part of the lesser sac as consisting of three subregions: retrogastric, perigastric and greater omental (Fig. 60.7C(ii)). The names are self-explanatory but their confines are all modified by various factors. Two phenomena are particularly prominent: namely, gastric 'descent' relative to the liver, and fusion of peritoneal layers with altered lines of reflection, adhesion of surfaces and loss of parts of cavities.

After the third month, hepatic growth, particularly of the left lobe, diminishes and the whole organ recedes into the upper abdomen. Meanwhile, the stomach elongates and some descent occurs, despite its relatively fixed cranial and caudal ends. This produces the angular flexure of the stomach, which persists postnatally. The concavity of the lesser curvature is now directed more precisely to the right; the lesser omentum is more exactly coronal and its free border vertical. Ventral to the liver, the free border of the falciform ligament passes steeply craniodorsally from umbilicus to liver (see disposition in the neonate in Figs 14.6–14.7). The mesenchymal dorsal wall of the lower part of the lesser sac, which is crossed obliquely by the growing pancreas, has, up to this point, remained free and retained its original dorsal midline root. Substantial areas now fuse with adjacent peritonealized surfaces of retroperitoneal viscera, the parietes, or another mesenteric sheet or fold. Where sheets fuse, there is a variable loss of apposed mesothelia and some continuity of their mesenchymal cores, but they remain surgically separable and no vascular anastomosis develops across the interzone. Above the pancreas, the posterior secondary dorsomesogastric wall of the sac becomes closely applied to the peritoneum covering the posterior abdominal wall and its sessile organs, the diaphragm, much of the left suprarenal gland, the ventromedial part of the upper pole of the left kidney, the initial part of the abdominal aorta, the coeliac trunk and its branches, and other vessels, nerves and lymphatics. Their peritoneal surfaces fuse. However, albeit with some tissue loss, a single mesothelium remains covering these structures, intercalated as a new secondary dorsal wall for this part of the lesser sac.

The pancreas grows from the duodenal loop, penetrating the substance of the dorsal mesoduodenum and secondary dorsal mesogastrium; their mesenchymes and mesothelia initially clothe its whole surface, except where peritoneal lines of reflection exist. Its posterior peritoneum becomes closely applied to that covering all the posterior abdominal wall structures it crosses (the inferior vena cava, abdominal aorta, splenic vein, superior mesenteric vessels, inferior mesenteric

vein, portal vein, left renal vessels, the caudal pole of the left suprarenal gland, a broad ventral band on the left kidney and various muscles). The intervening peritoneal mesothelia fuse and atrophy, and the mesenchymal cores form fascial sheaths and septa. The pancreas is now sessile. The peritoneum covering the upper left part of its head, neck and the anterosuperior part of its body forms the central part of the dorsal wall of the lesser sac. The pancreatic tail remains peritonealized by a persisting part of the secondary dorsal mesogastrium as it curves from the ventral aspect of the left kidney towards the hilum of the spleen. The infracolic parts of the pancreas are covered with greater sac peritoneum. In the greater omental subregion of the lower part of the lesser sac, two contrasting forms of mesenteric adhesion occur. The posterior 'returning' bilaminar stratum of the omentum undergoes partial fusion with the peritoneum of the transverse colon at the taenia omentalis and with its mesocolon. The layers remain surgically separable; no anastomosis occurs between omental and colic vessels.

The original dorsal midline attachment to the parietes of the foregut dorsal mesentery is profoundly altered during the development of the lesser sac and its associated viscera. However, despite the extensive areas of fusion, virtually the whole of the gastric greater curvature (other than a small suboesophageal area) and its topographical continuation (the inferior border of the first 2–3 cm of the duodenum) retain true mesenteric derivatives of the secondary dorsal mesogastrium and its continuation, the dorsal mesoduodenum. Although regional names are used to assist identification and description, it is important to emphasize that they are all merely subregions of one continuous sheet.

The upper (oesophagophrenic) part of the lesser omentum arches across the diaphragm. As this bilaminar mesentery approaches the oesophageal hiatus, its laminae diverge, skirting the margins of the hiatus. They then descend for a limited distance and with variable inclination, to enclose reciprocally shaped areas on the dorsum of the gastric fundus and diaphragm. The area may be roughly triangular to quadrangular; it contains areolar tissue and constitutes the bare area of the stomach or, when large, the left extraperitoneal space. Its right lower angle is the base of the left gastropancreatic fold, and its left lower angle reconstitutes the bilaminar mesentery. The root of the latter arches downwards and to the left across the diaphragm and suprarenal gland, and gives the gastrophrenic ligament to the gastric fundus. It continues to arch across the ventral surface of the upper part of the left kidney, and its layers part to receive the pancreatic tail; they initially extend to the hilum of the spleen as the splenorenal ligament (see Figs 60.6C, 60.7D). The left half of this bilaminar 'ligament' provides an almost complete peritoneal tunic for the spleen. It then reunites with its fellow at the opposite rim of the splenic hilum, and continues to the next part of the gastric greater curvature as the gastrosplenic ligament. The remaining part (perhaps two-thirds) of the gastric greater curvature and its short duodenal extension provide attachment for the anterior, 'descending', bilaminar stratum of the greater omentum. Its returning, posterior, bilaminar stratum continues to its parietal root (which extends from the inferior limit assigned to the splenorenal ligament), and curves caudally and to the right along the anterior border of the body of the pancreas, immediately cranial to the line of attachment of the transverse mesocolon. Crossing the neck of the pancreas, the same curve is followed for a few centimetres on to its head; the omental root then sharply recurves cranially and to the left, to reach the inferior border of the duodenum. Thus, it reaches that part of the lesser sac provided by cleavage of the dorsal mesoduodenum from the greater sac. It enters the epiploic foramen, traverses the epiploic canal between the caudate hepatic process and proximal duodenum, then crosses the right gastropancreatic fold, and descends behind the proximal duodenum to enter the right marginal strip enclosed by the greater omentum. The definitive origins of the peritoneum from the posterior abdominal wall are shown in the adult in Figures 63.1, 68.5, 63.6B.

Peritoneum associated with the mid- and hindgut

It is convenient to consider the mesenteries of the small and large intestine after rotation and the principal growth patterns have been achieved, and the developing pancreas is becoming retroperitoneal.

Small intestine

Most of the duodenal loop encircles the head of the pancreas and is retroperitoneal. The peritoneum principally covers its ventral and

convex aspects. Areas not covered are a short initial segment of the superior (first) part, which is more completely peritonealized because it gives attachment to the right margins of the greater and lesser omenta; the sites where the transverse colon is closely apposed to the descending (second) part, or where the latter is crossed by the root of the transverse mesocolon; and the sites where the mesentery crosses the transverse (third) part, and descends across the ascending (fourth) part from its upper extremity at the duodenojejunal flexure. These regions are illustrated in the adult in [Figure 63.2](#). In addition, one or more of up to six different duodenal recesses may develop ([p. 1108](#)).

From a mesenteric standpoint, the succeeding small intestine, from the duodenojejunal flexure to the ileocaecal junction, undergoes less modification of its embryonic form than other gut regions. Its early dorsal mesentery is a continuous, single (but structurally bilaminar) sheet, with a midline parietal attachment (line of reflection, or 'root'). The attachment of the root becomes an oblique narrow band from the left aspect of the second lumbar vertebra to the cranial aspect of the right sacroiliac joint (see [Figs 60.6H, 63.2](#)).

Ascending colon

The caecum and vermiform appendix arise as a diverticulum from the antimesenteric border of the caudal limb of the midgut loop; consequently, the caecum does not possess a primitive mesocaecum. These regions of the gut undergo long periods of growth, often asymmetrical, and their final positions, dimensions and general topography show considerable variation. The vermiform appendix is almost wholly clothed with visceral peritoneum, derived from the diverging layers of its rather diminutive mesoappendix. The latter should perhaps be regarded as a direct derivative of the primitive dorsal mesentery, and perhaps a similar status for the vascular fold of the caecum should be considered.

The colonic gut retains its primitive dorsal mesentery, the mesocolon, until the differential growth, rotation and circumabdominal displacement of this part of the gut tube near completion. Its original root is still vertical in the dorsal midline, although the mesocolon diverges from it widely as an incomplete, flattened pyramid, to reach its colonic border at the future taenia mesocolica. During the fourth and fifth months, substantial areas of the primitive mesocolon adhere to, then fuse with, the parietal peritoneum. In this way, some colonic segments become sessile while others have a shorter mesocolon with an often profoundly altered parietal line of attachment. The mesocolon of the transverse and sigmoid segments normally persists, while the ascending colon, right (hepatic) flexure and descending colon become sessile; the ascending or descending, or both, colonic segments may also retain a mesocolon, which varies from a localized 'fold' to a complete mesocolon. When sessile, the ventral, medial and lateral aspects of the ascending or descending colon are clothed with peritoneum, and the protrusion of the viscus produces medial and lateral peritoneal paracolic gutters on each side. This form of apposition to underlying structures (zygosis) proceeds from the ascending colon to include the right colic (hepatic) flexure; from there, it continues anteroinferiorly to the left, so involving the right-sided initial segment of the transverse colon.

Transverse colon

The right extremity of the transverse colon is sessile, and is separated by fibroareolar tissue from the anterior aspect of the descending (second) part of the duodenum and the corresponding aspect of most of the head of the pancreas. The remainder of the transverse colon, up to and including the left (splenic) colic flexure, is almost completely peritonealized by the diverging layers of the transverse mesocolon. The root of the latter reaches the neck and whole extent of the anterior border of the body of the pancreas. The long axis of the definitive pancreas lies obliquely. The splenic colonic flexure is considerably more rostral than the hepatic flexure and, consequently, the root of the mesocolon curves obliquely upwards as it crosses the upper abdomen from right to left. As it expands, the posteroinferior wall of the greater omental part of the lesser sac gradually covers, and becomes closely applied to, the transverse mesocolon and its contained colon, finally projecting beyond the latter. Craniocaudal adherence now occurs between the omental wall and the pericolonic and mesocolonic layers.

Descending colon

The left colic flexure receives much of its peritoneal covering from the left extremity of the transverse mesocolon. It is also often connected to the parietal peritoneum of the diaphragm over the tenth and

eleventh ribs by a phrenicocolic ligament. The latter sometimes blends with a presplenic fold that radiates from the gastrosplenic ligament. The descending colon becomes sessile. The process of fusion and obliteration of both ascending and descending mesocolons starts laterally and progresses medially.

Sigmoid colon

The sigmoid colon is most variable in its length and disposition. It retains its dorsal mesocolon, but the initial midline dorsal attachment of its root is considerably modified in its definitive state.

Rectum

The rectum continues from the ventral aspect of the third sacral vertebra to its anorectal (perineal) flexure anteroinferiorly to the tip of the coccyx; the distance changes with age. All aspects of the rectum are encased by mesenchyme, and the early, dorsally placed, mass is named, by some authorities, the dorsal mesorectum. However, the latter does not form a true mesentery; with progressive skeletal development, it is reduced to a woven fibroareolar sheet that displays patterned variations in thickness and fibre orientation. The sheet is closely applied to the ventral concavity of the sacrum and coccyx, and encloses numerous fibromuscular and neurovascular elements. The rectum, therefore, becomes sessile, and visceral peritoneum is restricted to its lateral and ventral surfaces (see [Fig. 72.4](#)).

With the disappearance of the postanal gut by the end of the fifth week, the ventrolateral peritoneum reaches the superior surface of the pelvic floor musculature; this condition persists until late in the fourth month. In the male, the ventral rectal peritoneum is reflected over the posterior surface of the prostate, bladder trigone and associated structures. In the female, the ventral peritoneum initially receives a reflection that covers almost the whole posterior aspect of the vagina, and is continued over the uterus. Subsequently, the closely apposed walls of these deep peritoneal pouches fuse over much of their caudal extent, their mesothelia are lost, and the viscera are separated by an intervening, bilaminar (surgically separable), fibrous stratum. In the male, this becomes the rectovesical fascia and posterior wall of the prostatic sheath (see [Fig. 72.10](#)). In the female, it becomes the rectovaginal septum between the lower part of the vagina and the rectum (see [Fig. 72.9](#)). The proximal third of the rectum is covered by peritoneum ventrolaterally; the lateral extensions of this tunic are triangular and deep proximally, but taper to an acute angle by the middle third of the rectum. The middle third of the rectum is covered by peritoneum only on its ventral surface, where it forms the posterior wall of the shallower rectovesical or rectovagino-uterine pouch. The remaining rectum and anal canal are extraperitoneal.

NEONATAL PERITONEAL CAVITY

The fully formed peritoneal cavity, although complex topographically, remains a single cavity with numerous intercommunicating regions, pouches and recesses (see [Fig. 14.7](#)). The only small peritoneal sacs to separate completely from the main cavity are the infracardiac bursa and the tunica vaginalis testis (see [Fig. 72.17](#)).

In fetal life, the greater omental cavity extends to the internal aspect of the lateral and caudal edges of the omentum. Postnatally, a slow but progressive fusion of the internal surfaces occurs, with obliteration of the most dependent part of the cavity; this proceeds rostrally and, when mature, the cavity does not usually extend appreciably beyond the transverse colon. Transverse mesocolon–greater omentum fusion begins early while the umbilical hernia of the midgut has not returned. It starts between the right margin of the early greater omentum and near the root of the presumptive mesocolon, and later spreads to the left.

In the neonate, the peritoneal cavity is ovoid (see [Fig. 14.6](#)). It is fairly shallow from anterior to posterior because the bilateral posterior extensions on each side of the vertebral column, which are prominent in the adult, are not present. Two factors lead to the protuberance of the anterior abdominal wall in the neonate and infant. The diaphragm is flatter in the newborn, which produces a caudal displacement of the viscera. The pelvic cavity is very small in the neonate, which means that organs that are normally pelvic in the adult, i.e. urinary bladder, ovaries and uterus, all extend superiorly into the abdomen (see [Figs 14.7, 72.9–72.10](#)). The pelvic cavity is joined to the abdominal cavity at less of an acute angle in the neonate because there is no lumbar vertebral curve and only a slight sacral curve.

The peritoneal attachments are similar to the adult. However, the greater omentum is relatively small; its constituent layers of peritoneum

may not be completely fused, and it does not extend much below the level of the umbilicus. Generally, the length of the mesentery of the small intestine and of the transverse and sigmoid mesocolons is greater than in the adult, whereas the area of attachment of the ascending and descending colons is relatively smaller. The peritoneal mesenteries and omenta contain little fat.

SPLEEN

The spleen appears about the sixth week as a localized thickening of the coelomic epithelium of the dorsal mesogastrium near its cranial end (see Figs 60.6–60.7). The proliferating cells invade the underlying angiogenic mesenchyme, which becomes condensed and vascularized. The process occurs simultaneously in several adjoining areas, which soon fuse to form a lobulated spleen of dual origin (from coelomic epithelium and from mesenchyme of the dorsal mesogastrium).

The vascular reticulum is well developed at 8–9 weeks, and contains immature reticulocytes and numerous closely spaced, thin-walled, vascular loops. From 11 weeks, the development of the spleen has been described in four stages: stage 0 has erythrocyte precursors, few macrophages and fewer lymphocytes; at stage I, lymphocytes begin colonization and arterial vascular lobules are seen; at stage II, T and B lymphocytes form peri-arteriolar clusters with B cells aggregated around the peripheral branches of splenic arterioles and T cells centrally; at stage III, before birth, the lobular arrangement is no longer discernible (Steiniger et al 2007).

From week 17, α -SMA-positive reticulum cells are found around arterioles. These increase in number and form a reticular framework from 20 to 23 weeks, when a primitive white pulp can be observed around arterioles. By 24 weeks, T and B lymphocytes become segregated, with T cells within the SMA-positive reticular framework, and B cells aggregating close to the peri-arteriolar lymphoid sheath, where they form lymphoid follicles (Satoh et al 2009).

Initially, the splenic capsule consists of cuboidal cells bearing cilia and microvilli. The enlarging spleen projects to the left, so that its surfaces are covered by the peritoneum of the mesogastrium on its left aspect, which forms a boundary of the general extrabursal (greater) sac. When fusion occurs between the dorsal wall of the lesser sac and the dorsal parietal peritoneum, it does not extend to the left as far as the spleen, which remains connected to the dorsal abdominal wall by a short splenorenal ligament. Its original connection with the stomach persists as the gastrosplenic ligament. The earlier lobulated character of the spleen disappears, but is indicated by the presence of notches on its upper border in the adult.

The spleen displays various developmental anomalies, including complete agenesis, multiple spleens or polysplenia, isolated small additional spleniculi and persistent lobulation. Asplenia and polysplenia are associated with other anomalies, especially those involving the cardiac and pulmonary systems. Accessory spleens are very common in neonates, located in the greater omentum. At birth, the spleen weighs, on average, 13 g (see Fig. 14.6). It doubles its weight in the first postnatal year and triples it by the end of the third year.

SUPRARENAL GLANDS

The suprarenal (adrenal) cortex is formed during the second month by a proliferation of the coelomic epithelium. Cells pass into the underlying mesenchyme between the root of the dorsal mesogastrium and the mesonephros (see Fig. 17.11). The proliferating tissue, which extends from the level of the sixth to the twelfth thoracic segments, is soon disorganized dorsomedially by invasion of neural crest cells from somite levels 18–24, which form the medulla, and also by the development of venous sinusoids. The latter are joined by capillaries, which arise from adjacent mesonephric arteries and penetrate the cortex in a radial manner. When proliferation of the coelomic epithelium stops, the cortex is enveloped ventrally, and later dorsally, by a mesenchymal capsule that is derived from the mesonephros. The subcapsular nests of cortical cells are the rudiment of the zona glomerulosa; they proliferate cords of cells that pass deeply between the capillaries and sinusoids. The cells in these cords degenerate in an erratic fashion as they pass towards the medulla, becoming granular, eosinophilic and, ultimately, autolysed. These cords of cells constitute the fetal cortex, which undergoes rapid involution during the first 2 years after birth. The fascicular and reticular zones of the adult cortex are proliferated from the glomerular zone after birth. Serial ultrasound measurements of the length of fetal suprarenal glands at 4-week intervals from the fifteenth week of gestation have been published (van Vuuren et al 2012).

The most common anomaly of suprarenal gland development is congenital hyperplasia, which occurs in 1:5000 to 1:15,000 births. This condition is caused by a group of autosomal recessive disorders, in which there are deficiencies in enzymes required for the synthesis of cortisol. In 90% of cases, the cause is deficiency of the enzyme 21-hydroxylase, producing an accumulation of 17-hydroxyprogesterone, which is converted to androgens. The levels of androgens increase by several hundred times, causing female embryos and fetuses to undergo external genital masculinization ranging from clitoral hypertrophy to formation of a phallus and scrotum; masculinization of the brain has also been suggested. In male embryos, the levels do not cause any changes in external genitalia. Signs of androgen excess may appear in childhood with precocious masculinization and accelerated growth (Lewis et al 1999).

SUPRARENAL GLANDS IN THE NEONATE

The suprarenal glands are relatively very large at birth (see Figs 14.6C, 72.8) and constitute 0.2% of the entire body weight, compared with 0.01% in the adult. The left gland is heavier and larger than the right, as it is in the adult. At term, each gland usually weighs 4 g; the average weight of the two glands is 9 g (average in the adult is 7–12 g). Estimation of the suprarenal gland volume can provide an estimate of fetal weight; a volume of greater than 420 mm³/kg is relevant in predicting preterm birth within 5 days of measurement (Turan et al 2012, Turan et al 2007). Within the first 2 weeks of postnatal life, the glands shrink to normal infantile size. The average weight of both glands is 5 g by the end of the second week, and 4 g by 3 months; gland birth weight is not regained until puberty. This rapid involution of the glands occurs regardless of gestational age. It is thought that parturition is the stimulus for suprarenal involution (Ben-David et al 2007). The cortex of the suprarenal gland is thicker than in the adult and the medulla of the gland is small. With normal involution, the fetal zone cells of the postnatal gland become smaller and they assume the appearance and organization typical of zona fasciculata.

INFERIOR VENA CAVA, PORTAL CIRCULATION AND UMBILICAL VESSELS

INFERIOR VENA CAVA

The inferior vena cava of the adult is a composite vessel that develops on the posterior abdominal wall dorsal to the developing peritoneal cavity. It forms as a consequence of the temporal remodelling of successive venous complexes (see Fig. 13.4). The precise mode of development of its postrenal segment (caudal to the renal vein) is still somewhat uncertain. Its function is initially carried out by the right and left postcardinal veins (Fig. 60.8; see also Fig. 13.4), which receive the venous drainage of the lower limb buds and pelvis, and run in the dorsal part of the mesonephric ridges, receiving tributaries from the body wall (intersegmental veins) and from the derivatives of the mesonephroi. It should be remembered that descriptions of venous development are very largely based on studies on animals, where the dimensions and final disposition of visceral organs differ from those found in humans. Many of the changes seen in the development of the infrahepatic caval and azygos systems in the human may result from lateral to medial movement of the vessels as a consequence of the growth of the abdominal viscera (Hikspoors et al 2015).

The early postcardinal veins communicate across the midline via an interpostcardinal anastomosis. This remains as an oblique transverse anastomosis between the iliac veins, and becomes the major part of the definitive left common iliac vein. It diverts an increasing volume of blood into the right longitudinal veins, which accounts for the ultimate disappearance of most of those on the left.

The supracardinal veins receive the larger venous drainage of the growing body wall. The right supracardinal vein persists and forms the greater part of the postrenal segment of the inferior vena cava. The continuity of the vessel is maintained by the persistence of the anastomosis between the right supracardinal and the right subcardinal in the renal collar. The left supracardinal disappears, but some of the renal collar formed by the left supracardinal–subcardinal anastomosis persists in the left renal vein. Both supracardinal veins drain cranially into the subcardinal veins. On the right side, the subcardinal vein comes into intimate relationship with the liver. An extension of the vessel takes place in a cranial direction and meets and establishes continuity with a corresponding new formation that is growing caudally from the right

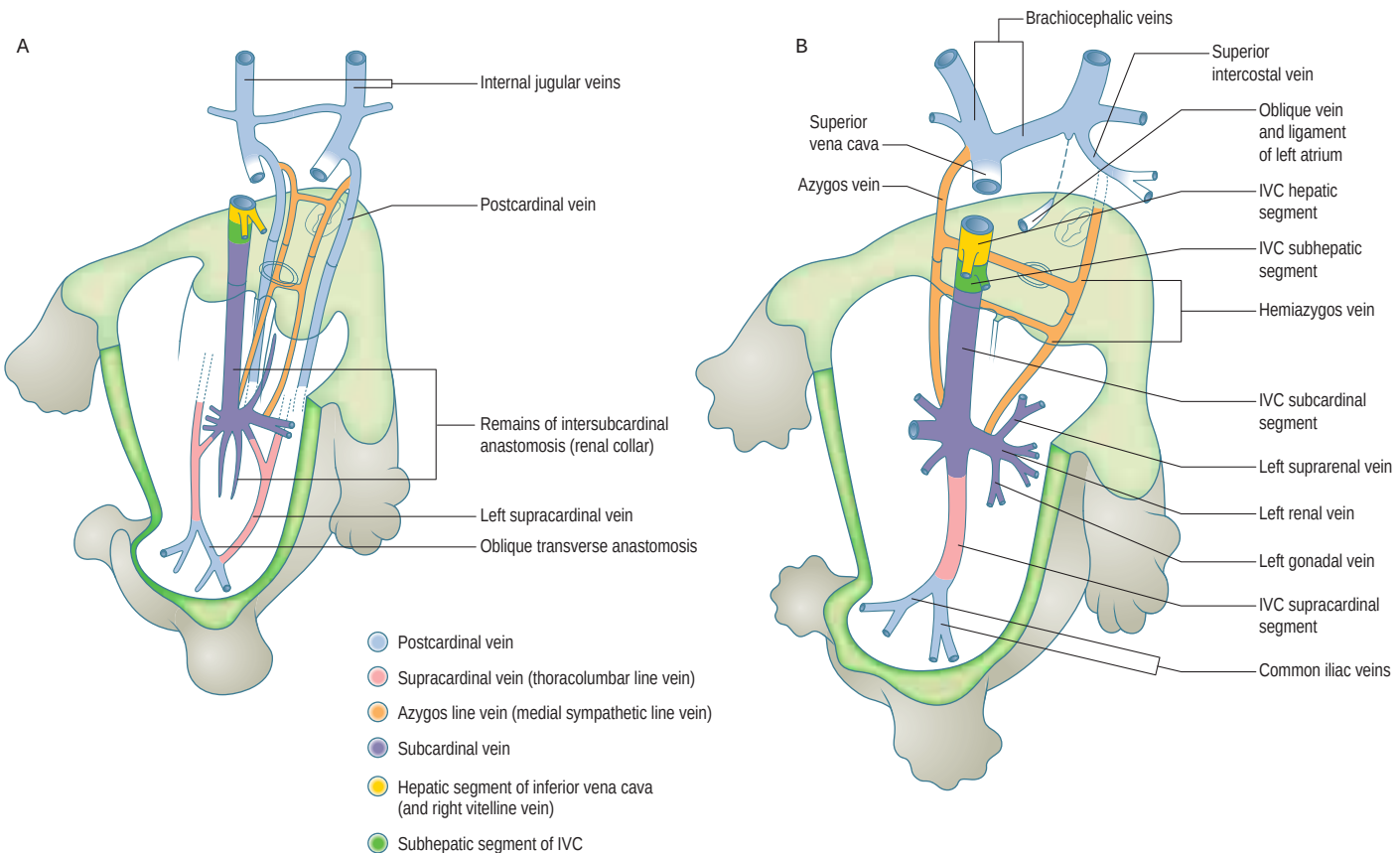


Fig. 60.8 The inferior vena cava (IVC) develops within the posterior abdominal wall. It is a composite vessel formed by temporal remodelling of successive somatic venous anastomoses. **A**, The progressive asymmetry of the developing venous system. **B**, The definitive venous arrangement. (For early venous development, see Fig. 13.4.)

vitelline hepatocardiac (common hepatic) vein. In this way, on the right side, a more direct route is established to the heart and the prerenal (cranial) segment of the inferior vena. In summary, the inferior vena cava is formed from below upwards by the confluence of: the common iliac veins, a short segment of the right postcardinal vein, the postcardinal–supracardinal anastomosis, part of the right supracardinal vein, the right supracardinal–subcardinal anastomosis, right subcardinal vein, a new anastomotic channel of double origin (the hepatic segment of the inferior vena cava), and the cardiac termination of the right vitelline hepatocardiac vein (common hepatic vein).

Only the supracardinal part of the inferior vena cava receives intersegmental venous drainage. The postrenal (caudal) segment of the inferior vena cava is on a plane that lies dorsal to the plane of the prerenal (cranial) segment. Thus, the right phrenic, suprarenal and renal arteries, which represent persistent mesonephric arteries, pass behind the inferior vena cava, whereas the testicular or ovarian artery, which has a similar developmental origin, passes anterior to it.

In some animals, the right postcardinal vein constitutes a large part of the postrenal segment of the inferior vena cava. In these cases, the right ureter, on leaving the kidney, passes medially dorsal to the vessel and then, curving round its medial side, crosses its ventral aspect. Rarely, a similar condition is found in humans, and indicates the persistence of the right postcardinal vein and failure of the right supracardinal to play its normal part in the development of the inferior vena cava.

PORTAL CIRCULATION

The heart first differentiates in splanchnopleure that, after head-fold formation, forms the dorsal wall of the primitive pericardial cavity (floor of the rostral foregut) and may, therefore, be considered a highly specialized vitelline vascular derivative. At the caudal extremity of the splanchnopleuric gut tube (the future lower rectum and upper anal canal), the vitelline venous drainage makes connections with the internal iliac radicles of the postcardinal complex.

The development of the gut occurs contemporaneously with changes to the early symmetrical embryonic circulation (Ch. 13). A symmetrical, segmental system of somatic arteries remains and supplies the body wall of the trunk. The underlying arrangement of these vessels is only slightly modified by subsequent development. The main changes to the

embryonic arrangement occur with the growth of the foregut, liver and paired viscera, i.e. kidneys and suprarenal glands.

The early development of the ventral splanchnic arteries is outlined in Chapter 13. Three main arterial trunks – the coeliac trunk, superior mesenteric and inferior mesenteric arteries – supply the fore-, mid- and hindgut, respectively. From the gastric terminal segment of the future oesophagus to the upper rectum, the developing endodermal epithelium is surrounded by splanchnopleuric mesenchyme permeated by a capillary plexus that drains into an anastomosing network of veins. This plexus is particularly dense ventrally where it is associated with the central midgut region; here, for a while, it receives a network of small veins from the definitive yolk sac. Later, in normal development, these vessels atrophy as the yolk sac diminishes. More rostrally, the splanchnopleuric capillaries associated with the foregut and upper portions of the yolk sac form longitudinal channels anterolateral to the gut and these become increasingly well defined as the embryonic abdominal vitelline veins. Entering the septum transversum, the right and left vitelline veins incline slightly, becoming parallel to the lateral aspects of the gut. They establish connections with capillary plexuses in the septal mesenchyme, and then continue, finally curving to enter the medial part of the cardiac sinusal horn of their corresponding side. The parts of the gut closely related to the presinual segments of the vitelline veins are the future subdiaphragmatic end of the oesophagus, primitive stomach, the superior (first) and descending (second) regions of the duodenum, and the remainder of the duodenal tube.

The principal ascending vitelline veins flanking the sides of the abdominal part of the foregut receive venules from its splanchnopleuric capillaries and those of the septal mesenchyme. Within these venular nets, enlarged (but still plexiform) anastomoses connect the two vitelline veins (Fig. 60.9; see Fig. 13.1C). A subdiaphragmatic intervittelline anastomosis develops in the rostral septal mesenchyme, lying a little caudal to the cardiac sinuatrial chamber, connecting the veins near their sinual terminations. (The channel is sometimes termed suprahepatic because of the position of the hepatic primordium; with expansion of the latter, it becomes partly intrahepatic.) The presumptive duodenum is crossed by three transverse duodenal intervittelline anastomoses. Their relation to the gut tube alternates so that the most cranial, the subhepatic, is ventral, the intermediate is dorsal and the caudal is ventral (Fig. 60.9B). It has become customary to describe the paraduodenal vitelline veins and their associated anastomoses as forming a figure eight. At this early stage, when left and right embryonic veins are still symmetrical,

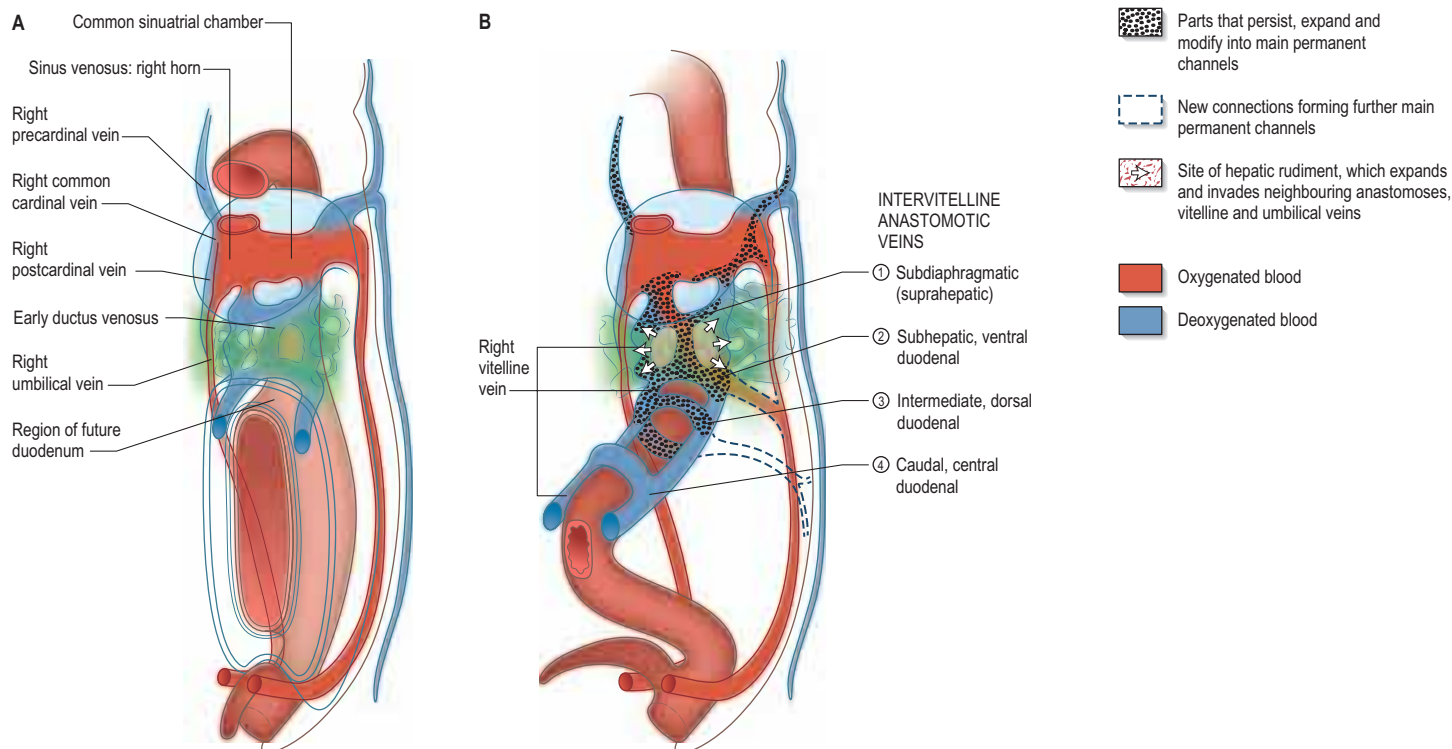


Fig. 60.9 Development of the vitelline, umbilical and terminal cardinal vein complexes: the early symmetrical condition. **A**, The early symmetrical arrangement of cardinal and umbilical veins entering the sinus venosus; blood from the yolk sac passes, via vitelline veins, through anastomoses forming in the septum transversum mesenchyme (green). **B**, The developing duodenum is surrounded by transverse duodenal intervittelline anastomoses. Changes in the developing gut and umbilical veins cause shifts in the routes taken by venous blood flowing to the sinus venosus.

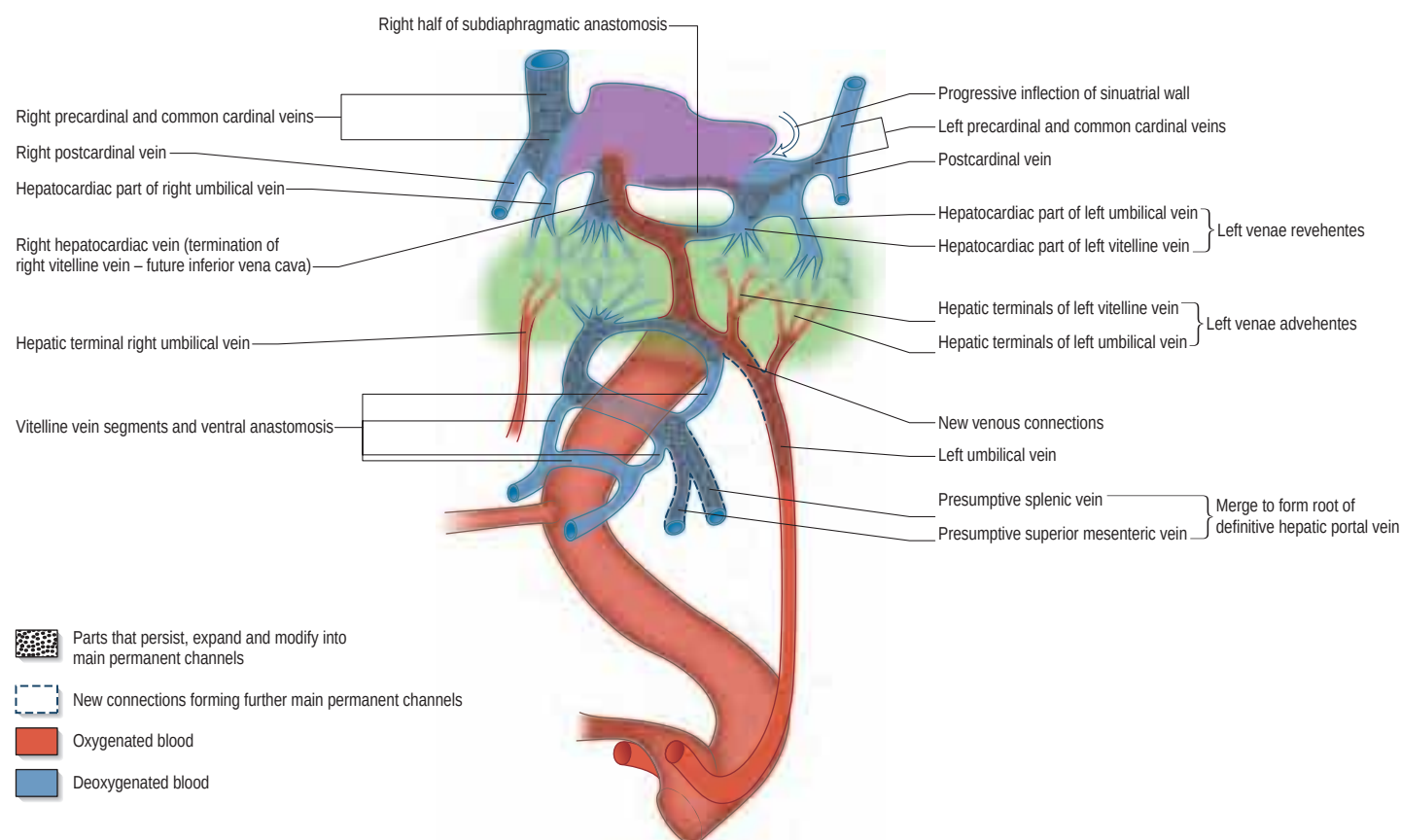


Fig. 60.10 Development of the vitelline, umbilical and terminal cardinal vein complexes: a mid-stage of asymmetry has been reached between the early symmetrical condition (see Fig. 60.9) and the definitive late prenatal state. The extent of the liver within the septum transversum mesenchyme is shown in green.

the cranial duodenal anastomosis becomes connected with the subdiaphragmatic anastomosis by a median longitudinal channel, the primitive median ductus venosus, which is dorsal to the expanding hepatic primordium but ventral to the gut. The further development of the vitelline veins and anastomoses is, as indicated, closely interlocked, with rapid hepatic expansion and gut changes, and umbilical vein disposition and modification is closely involved.

CHANGES IN THE VITELLO-UMBILICAL VEINS

Progressive changes in the vitello-umbilical veins are rapid, profound and closely linked with regional modifications in shape and position of the gut, expansion and invasion of venous channels by hepatic tissue, asymmetry of the heart and cardiac venous return. The principal events are summarized in Figures 60.9–60.11.

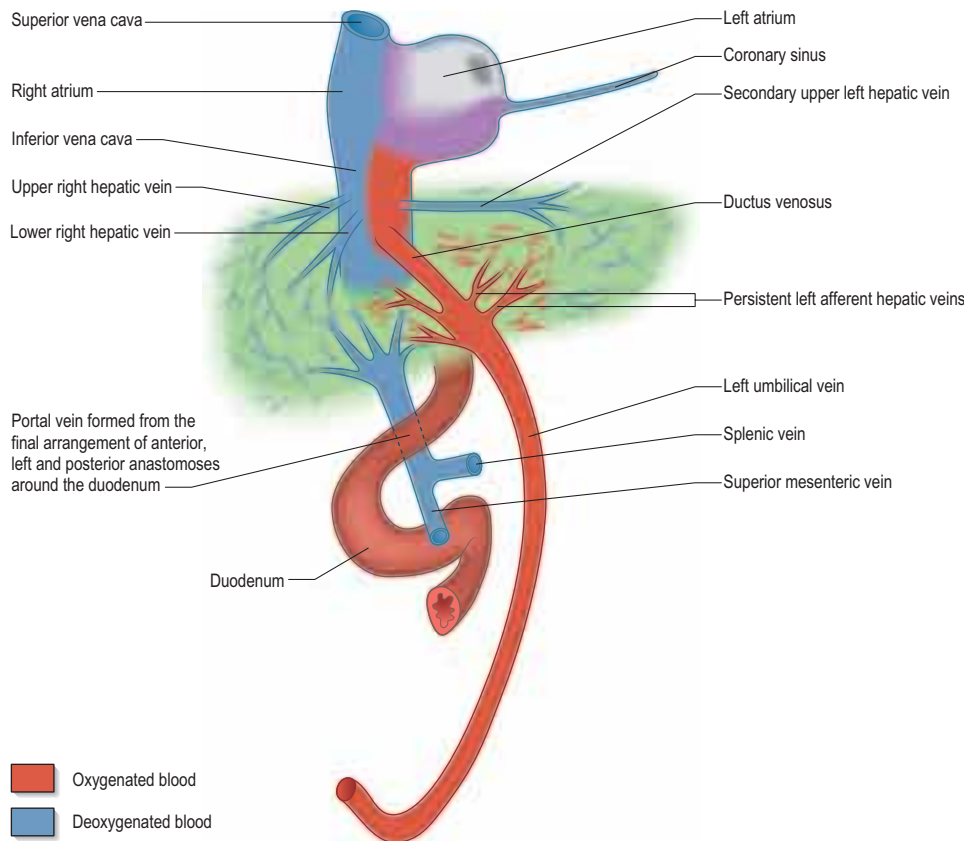


Fig. 60.11 The condition of some main upper abdominal and intrathoracic right atrial terminal veins in the later prenatal months. The rotation of the stomach and movement of the duodenum produce relative changes in the final arrangement of the duodenal intervilline anastomoses that form the hepatic portal vein.

From the cranial portion of the early hepatic evagination of the foregut, interconnected sheets and 'cords' of endodermal cells, the presumptive hepatocytes, penetrate the septum transversum mesenchyme. Hepatic mesenchyme is composed of a mixed population of cells with endothelial/angiogenic and connective tissue lineages. Capillary plexuses develop within this mesenchyme and, possibly under the influence of the endodermal hepatic sheets, the plexuses become more profuse by further addition of angioblastic septal mesenchyme, which also forms masses of perivascular intrahepatic haemopoietic tissue. These processes extend along the plexiform connections of the vitelline, and later the umbilical, veins until their intrahepatic (trans-septal) zones themselves become largely plexiform. Initially capillary in nature, they transform into a mass of rather wider, irregular, sinusoidal vessels with a discontinuous endothelium containing many phagocytic cells. The lengths of vitelline veins involved in these processes are the intermediate parts of the segments extending from the subhepatic (cranioventral duodenal) to the suprahepatic (subdiaphragmatic) transverse intervilline anastomoses, and the corresponding lengths of the umbilical veins. Thus, at this early stage, the liver sinusoids are perfused by mixed blood reaching them through a series of branching vessels collectively called the *venae advehentes*, or afferent hepatic veins: they are deoxygenated from the gut splanchnopleure via vitelline vein hepatic terminals, and oxygenated from the placenta via hepatic terminals of the umbilical veins (see Fig. 60.10). Blood leaves the liver through four *venae revehentes* (efferent hepatic veins); two on each side reach and open into their respective cardiac sinusal horns. This full complement of four hepatocardiac veins is only transient and becomes reduced to one dominant, rapidly enlarging channel. The originally bilaterally symmetrical cardinal vein complexes, both rostral and caudal, develop transverse or oblique anastomoses so that the cardiac venous return is ultimately restricted to the definitive right atrium.

These cardiac and concomitant hepatocentric changes are accompanied by events in supra-, intra- and subhepatic parts of the vitello-umbilical veins. Some vessels enlarge, persisting as definitive vessels to maturity; in places, they are joined later by other channels that become defined in already established capillary plexuses. Other vessels retrogress, either disappearing completely or remaining as vestigial tags and, occasionally, vessels of fine calibre. Some vessels of crucial importance in the circulatory patterns of embryonic and fetal life become obliterated postnatally and transformed to substantial fibrous cords. Both right and left umbilical hepatocardiac and the left vitelline hepatocardiac veins continue, for a time, to discharge blood into their sinusal horns; however, they begin to retrogress (see Figs 60.9–60.10). The right umbilical channel atrophies completely. The left channels also disappear, but their cardiac terminals may, on occasion, be found as conical

fibrous tags attached to the inferior wall of the coronary sinus. The right vitelline hepatocardiac vein continues enlarging and, ultimately, forms the terminal segment of the inferior vena cava. The latter receives the right efferent hepatic veins and new channels draining the territories of the left efferent hepatic veins. These collectively form the upper and lower groups of right and (secondary) left hepatic veins. The terminal caval segment also shows the orifice of the right half of the intervilline subdiaphragmatic anastomosis, and a large new connection with the right subcardinal vein.

The hepatic terminals of the right and left duodenal parts of the vitelline veins are destined to form the corresponding branches of the hepatic portal vein, the left branch incorporating the cranial ventral intervilline anastomosis. With rotation of the gut and formation of the duodenal loop, segments of the original vitelline veins and the caudal transverse anastomosis atrophy (indicated in Figs 60.10–60.11), while new splanchnopleuric venous channels – the superior mesenteric and splenic veins – converge and join the left end of the dorsal intermediate anastomosis. The numerous other radicles of the portal vein and its principal branches, including the inferior mesenteric vein, are later formations.

For a period, placental blood returns from the umbilicus via right and left umbilical veins, both discharging through afferent hepatic veins into the hepatic sinusoids, where admixture with vitelline blood occurs. At approximately 7 mm crown-rump length, the right umbilical vein retrogresses completely. The left umbilical vein retains some vessels discharging directly into the sinusoids, but new enlarging connections with the left half of the subhepatic intervilline anastomosis emerge. The latter is the start of a bypass channel for the majority of the placental blood, which continues through the median ductus venosus and, finally, the right half of the subdiaphragmatic anastomosis, to reach the termination of the inferior vena cava (see Fig. 60.11).

The left and right hepatic portal veins and the ductus venosus are routinely observed in the second trimester ultrasound scan at the plane to estimate abdominal circumference (see Fig. 14.4G).

FETAL/NEONATAL UMBILICAL VESSELS

Umbilical arteries

The fetal umbilical arteries are in direct continuation with the internal iliac arteries. Their lumen is 2–3 mm in diameter at their origin, when distended. This narrows as they approach the umbilicus, where there is a reciprocal thickening of the tunica media, as a result, in particular, of an increase in the number of longitudinal smooth muscle fibres and

Table 60.1 Key anatomical reference points for umbilical arterial catheterization

Structure	Vertebral level
Ductus arteriosus	T4–5
Coeliac artery	T12
Superior mesenteric artery	T12–L1
Renal artery	L1
Inferior mesenteric artery	L3
Aortic bifurcation	L4–5

elastic fibres. Before birth, there is a proliferation of connective tissue within the vessel wall. The umbilical vessels constrict in response to handling, stretching, cooling and altered tensions of oxygen and carbon dioxide. Umbilical vessels are muscular, but devoid of a nerve supply in their extra-abdominal course. After the cord is severed, the umbilical arteries contract, preventing significant blood loss; thrombi often form in the distal ends of the arteries. The arteries obliterate from their distal ends until, by the end of the second or third postnatal month, involution has occurred at the level of the superior vesical arteries. The proximal parts of the obliterated vessels remain as the medial umbilical ligaments.

Umbilical arterial catheterization

Insertion of an umbilical catheter is undertaken to provide direct access to the arterial circulation. Arterial blood can be withdrawn repeatedly for measurement of oxygen and carbon dioxide partial pressures, pH, base excess and many other parameters of blood biochemistry and haematology. The indwelling catheter also enables the continuous measurement of arterial blood pressure.

The catheter is inserted directly into either the cut end or the side of one of the two umbilical arteries in the umbilical cord stump that remains attached to the baby after transection of the umbilical cord at the time of delivery. The catheter tip is then advanced along the length of the umbilical artery, through the internal iliac artery, into the common iliac artery and, from there, into the aorta. In order to keep the catheter patent, a small volume of fluid is continuously infused through it. It is important for the tip of the catheter to be located well away from arteries branching from the aorta, to avoid potentially harmful perfusion of these arteries with the catheter fluid. Thus, umbilical arterial catheter tips are placed in the descending aorta either in a ‘high’ position, above the coeliac artery but well below the ductus arteriosus, or in a ‘low’ position, below the renal and inferior mesenteric arteries but above the point where the aorta bifurcates into the two common iliac arteries. The length of catheter to be inserted can be estimated from charts relating the required catheter length to external body measurements, or from birth weight. Positioning of the catheter is assessed by means of radiographs of the abdomen or chest: a ‘high’ catheter tip should be located in the descending aorta somewhere between the levels of the sixth and ninth thoracic vertebrae (T6–9), while a ‘low’ catheter tip should be at a level between the third and fourth lumbar vertebrae (L3–4). Relevant anatomical reference points are given in Table 60.1.

Umbilical vein

The umbilical vein in the neonate is 2–3 cm long and 4–5 mm in diameter when distended. It passes from the umbilicus, within the layers of the falciform ligament, superiorly and to the right, to the porta hepatis. Here, it gives off several large intrahepatic branches to the liver and then joins the left branch of the portal vein and the ductus venosus. The umbilical vein is thin-walled; it possesses a definite internal lamina of elastic fibres at the umbilical ring but not in its intra-abdominal course. The tunica media contains smooth muscle fibres, collagen and elastic fibres. When the cord is severed, the umbilical vein contracts, but not so vigorously as the arteries. The rapid decrease in pressure in the umbilical vein after the cord is clamped means that the elastic tissue at the umbilical ring is sufficient to arrest any retrograde flow along the vessel. Before birth, there is a subintimal proliferation of connective tissue. After birth, the contraction of the collagen fibres in the tunica media and the increased subintimal connective tissue contribute to the transformation of the vessel into the ligamentum teres. Obliteration occurs from the umbilical ring towards the hepatic end. No thrombi are formed in the obliteration process. For up to 48 hours after birth, the intra-abdominal portion of the umbilical vein can be easily dilated. In most adults, the original lumen of the vein persists through the ligamentum teres and can be dilated to 5–6 mm in diameter.

Ductus venosus

The ductus venosus is a direct continuation of the umbilical vein and arises from the left branch of the portal vein, directly opposite the termination of the umbilical vein. It passes for 2–3 cm within the layers of the lesser omentum, in a groove between the left lobe and caudate lobe of the liver, before terminating in either the inferior vena cava, or in the left hepatic vein immediately before it joins the inferior vena cava. The tunica media of the ductus venosus contains circularly arranged smooth muscle fibres, an abundant amount of elastic fibres and some connective tissue.

The ductus venosus is already closed in about one-third of newborn infants. It shuts down by an unknown mechanism. Obliteration begins at the portal vein end in the second postnatal week and progresses to the vena cava; the lumen has been completely obliterated by the second or third month after birth. The fibrous remnant of the ductus venosus is the ligamentum venosum.

Umbilical vein catheterization

The umbilical vein may be catheterized in the neonate to enable exchange and transfusion of blood, for central venous pressure measurement and, usually in an emergency, for vascular access. The catheter is inserted into the cut end of the umbilical vein and is advanced along the length of the vein, through the ductus venosus and into the inferior vena cava; the tip is placed between the ductus venosus and the right atrium. Positioning of the catheter tip is confirmed radiologically and it should be located just above the diaphragm at a point that is level with the ninth or tenth thoracic vertebra (T9/T10). As with umbilical arterial catheters, estimation of the required catheter length can be determined from standard charts.

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